PREDICTING INCIDENT DEPRESSION IN OLDER ADULTS USING INTERPRETABLE MACHINE LEARNING: AN ANALYSIS OF THE ENGLISH LONGITUDINAL STUDY OF AGEING

A study submitted in partial fulfilment
of the requirements for the degree of
MSc Data Science

at

THE UNIVERSITY OF SHEFFIELD

by

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Word Count: 11,432

August 2026

ABSTRACT

Background. Depression is a common mental health condition among older adults and is associated with reduced quality of life, increased disability and greater healthcare utilisation. Early identification of individuals at increased risk may support preventative interventions before symptoms become established. Although machine learning has increasingly been applied to mental health research, relatively few studies have examined incident depression using nationally representative longitudinal data alongside model interpretability.

Aims. This study aimed to predict incident depression among adults aged 50 years and older using longitudinal data from the English Longitudinal Study of Ageing (ELSA). Secondary aims were to compare the predictive contribution of sociodemographic, health, behavioural and social domains, evaluate whether temporal changes improved prediction, assess subgroup performance and identify the most influential predictors.

Methods. A quantitative longitudinal secondary analysis was conducted using Waves 7 (2014–2015) and 8 (2016–2017) of ELSA. Participants with depression at baseline were excluded, resulting in an analytical cohort of 6,702 individuals. Logistic Regression, Random Forest and XGBoost models were developed and evaluated using stratified five-fold cross-validation. Performance was assessed using the area under the receiver operating characteristic curve (AUROC), sensitivity, specificity, predictive values, F1 score and calibration. Model interpretability was examined using adjusted odds ratios, SHapley Additive exPlanations and permutation importance.

Results. Logistic Regression achieved the strongest overall predictive performance (AUROC = 0.759), outperforming Random Forest (AUROC = 0.726) and XGBoost (AUROC = 0.668). All models demonstrated high specificity but poor sensitivity in the context of the low incidence of depression within the analytical cohort. Loneliness, functional limitation, poor sleep, physical activity and financial hardship consistently emerged as the strongest predictors. Health and social domains achieved stronger discrimination than behavioural and sociodemographic domains. Temporal change variables did not improve prediction beyond baseline measurements, while performance varied modestly across age and sex subgroups.

Conclusions. Incident depression among older adults was predicted with moderate discrimination using longitudinal survey data, although sensitivity remained limited. Logistic Regression outperformed the more complex models, indicating that greater model complexity

offered no predictive advantage in this structured survey dataset. The findings highlight the importance of social and functional health factors and support interpretable models for population-level risk stratification. Future research should investigate threshold optimisation, methods for addressing class imbalance and external validation to improve clinical applicability.

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my supervisor, Professor Peter A. Bath, for his guidance and constructive feedback throughout this dissertation. His expertise and advice were invaluable in shaping the direction and quality of this research.

I am also grateful to the University of Sheffield for providing the knowledge, skills and learning environment that prepared me to undertake this project. The teaching I received throughout my master's degree was fundamental to the completion of this dissertation.

My heartfelt thanks go to my family for their unwavering support, encouragement and belief in me throughout my studies. Their reassurance motivated me to persevere during the more challenging stages of this project.

I would also like to acknowledge the English Longitudinal Study of Ageing research team for collecting and maintaining the longitudinal dataset used in this study. I am grateful to the UK Data Service for providing access to these data.

Finally, I would like to thank everyone who contributed to my academic journey and supported me throughout the completion of this dissertation.

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LIST OF ABBREVIATIONS

Abbreviation	Definition
AI	Artificial Intelligence
aOR	Adjusted Odds Ratio
AUROC	Area Under the Receiver Operating Characteristic Curve
CES-D	Centre for Epidemiologic Studies Depression Scale
CI	Confidence Interval
ELSA	English Longitudinal Study of Ageing
MAR	Missing At Random
NPV	Negative Predictive Value
OR	Odds Ratio
PPV	Positive Predictive Value
SD	Standard Deviation
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
VIF	Variance Inflation Factor

1 INTRODUCTION

1.1 BACKGROUND

Depression is one of the most common mental health conditions affecting older adults and represents a major public health challenge worldwide. According to the World Health Organization (2023), depression contributes substantially to disability, reduced quality of life and increased healthcare utilisation among older populations. As life expectancy continues to increase globally, understanding the factors associated with depression in later life has become an increasingly important research priority (Gianfredi et al., 2025).

Older adults experience a unique combination of biological, psychological and social changes that may increase their vulnerability to depression. Age-related physical health decline, the accumulation of chronic diseases, reduced mobility, bereavement, retirement, financial pressures and social isolation have all been associated with poorer mental health outcomes. Loneliness and functional limitations have received particular attention due to their strong association with depressive symptoms in later life (Conde-Sala et al., 2019). These factors often interact, reflecting the multifactorial nature of depression (England & Sim, 2009).

A distinction must be made between prevalent and incident depression. Prevalent depression refers to existing cases identified at a specific point in time, whereas incident depression refers to cases occurring among individuals who were free from depression at baseline. Although many studies have focused on prevalent depression, predicting incident depression is particularly valuable because it supports earlier identification of individuals at elevated risk and may enable preventative interventions before symptoms become established (National Institute of Mental Health, 2020).

Traditional statistical methods have played an important role in understanding depression risk factors. However, predicting future depression remains challenging because risk reflects sociodemographic, health, behavioural and social factors alongside broader biological and environmental influences (Geurgas et al., 2025). This complexity has contributed to growing interest in machine learning approaches to mental health prediction (Nickson et al., 2023).

Machine learning algorithms identify patterns within data to generate predictions and can model non-linear relationships and interactions without requiring them to be specified in advance (Cho et al., 2019). The increasing availability of large longitudinal datasets has

expanded opportunities to apply these methods to population health research and depression prediction among older adults (Arseneault et al., 2025).

1.2 Rationale for the Study

Population ageing is one of the most significant demographic changes occurring worldwide, with the global population aged 60 years and over expected to exceed 2.1 billion by 2050. This growth is expected to increase the burden of age-related health conditions, including depression, strengthening the need for effective approaches to identifying individuals at increased risk (Gianfredi et al., 2025).

Despite extensive research examining depression in older populations, several important gaps remain, which are explored in detail in Chapter 2. First, many studies investigate prevalent rather than incident depression. This limits understanding of the factors associated with depression onset because predictors and outcomes are often assessed concurrently. Investigating incident depression allows predictors to be measured before the outcome occurs, providing stronger evidence regarding future risk (National Institute of Mental Health, 2020).

Second, relatively few studies have applied interpretable machine learning methods to predict depression among older adults using nationally representative longitudinal datasets. Many studies prioritise predictive accuracy while providing limited explanation of how predictions are generated. This lack of transparency may restrict implementation within healthcare, where interpretability is important for supporting clinical decision-making (Lamba et al., 2026).

Third, limited research has compared the predictive contribution of different predictor domains within a single modelling framework. Sociodemographic, health, behavioural and social factors have all been linked to depression risk, yet uncertainty remains regarding their relative predictive importance. Comparing these domains may improve understanding of depression aetiology and inform future prevention strategies (Geurgas et al., 2025).

Finally, limited evidence exists regarding whether temporal changes in risk factors improve prediction beyond baseline measurements. Longitudinal data provide opportunities to examine these changes, but their additional predictive value remains unclear.

The English Longitudinal Study of Ageing (ELSA) provides an appropriate dataset for addressing these gaps. ELSA contains detailed information on demographic characteristics, socioeconomic circumstances, health, behavioural factors and social participation among adults aged 50 years and older living in England. Its breadth, national representativeness and

repeated assessments make it particularly suitable for investigating incident depression using interpretable machine learning methods (Steptoe et al., 2012).

1.3 Aim of the Study

The primary aim of this study was to predict incident depression in Wave 8 of ELSA among adults aged 50 years and older who were free from depression in Wave 7.

To achieve this aim, supervised machine learning models were developed using sociodemographic, health, behavioural and social variables measured during Wave 7. The study also sought to evaluate model performance, identify important predictors of depression risk, compare the predictive contribution of different predictor domains, investigate whether temporal changes in selected variables improved predictive performance, and assess performance across age and sex subgroups.

1.4 Research Questions

This study addressed the following primary research question:

Can incident depression in Wave 8 of ELSA be predicted from Wave 7 sociodemographic, health, behavioural and social variables using supervised machine learning methods?

To address this question, five secondary research questions were investigated:

1. Which predictor domain contributes most to depression prediction: sociodemographic, health, behavioural or social factors?
2. Do changes in variables between Waves 7 and 8 improve predictive performance compared with baseline measurements alone?
3. Does predictive performance vary across age and sex subgroups?
4. Which variables are most important in predicting incident depression?
5. How well do the predictive models perform in terms of discrimination and calibration?

1.5 Hypotheses

Based on the existing literature reviewed in Chapter 2, the following hypotheses were proposed:

H1: Machine learning models developed using Wave 7 sociodemographic, health, behavioural and social predictors will discriminate between participants who do and do not develop depression at Wave 8.

H2: The predictive contribution differs across the sociodemographic, health, behavioural and social domains.

H3: Including temporal changes in variables between Waves 7 and 8 improves predictive performance.

H4: Predictive performance differs across age and sex subgroups.

H5: Loneliness, functional limitation and age are among the most important predictors of incident depression.

1.6 Structure of the Dissertation

The remainder of this dissertation is organised into five chapters.

Chapter 2 critically reviews the literature relevant to the present study and establishes the research gaps. Chapter 3 then describes the methodology used to address these gaps. Chapter 4 presents the results of the analyses, which are interpreted in relation to the research questions, hypotheses and existing literature in Chapter 5. Finally, Chapter 6 draws together the findings and presents the conclusions and recommendations for future research.

1.7 Introduction Summary

This chapter introduced the background, rationale and objectives of the study and defined the research questions and hypotheses guiding the investigation. It established the importance of predicting incident depression among older adults and highlighted the need for interpretable machine learning approaches using longitudinal survey data. The following chapter critically reviews the relevant literature and establishes the research gaps addressed by this study.

2 LITERATURE REVIEW

Building on the context, rationale and research questions established in the previous chapter, this chapter critically reviews the literature relevant to the present study. It examines the epidemiology and theoretical foundations of depression in older adults, evidence relating to sociodemographic, health, behavioural and social predictors, machine learning and interpretability approaches, longitudinal prediction and previous research using ELSA.

2.1 LITERATURE SEARCH STRATEGY

A structured literature search was undertaken to identify evidence relating to depression and its prediction in older adults, including applications of machine learning in mental health research. PubMed, Scopus, PsycINFO and Google Scholar were searched because they cover biomedical, psychological and multidisciplinary research.

Search terms were organised around four core concepts: depression, older adults, longitudinal research and machine learning. Keywords included *depression*, *mental health*, *ageing*, *older adults*, *prediction*, *longitudinal* and *machine learning*. Phrase-based searches included *later-life depression*, *incident depression*, *depression prediction in older adults*, *machine learning mental health prediction*, *longitudinal depression modelling* and *ELSA depression studies*. Boolean operators were used to combine concepts, including searches such as *depression AND older adults AND longitudinal*.

The review primarily included peer-reviewed journal articles published within the last 10 to 15 years to reflect current evidence and methods. Earlier studies were retained where they introduced influential theoretical or methodological frameworks. Studies were included if they focused on adults aged 50 years or older, examined depression longitudinally or predictively, or evaluated machine learning approaches relevant to depression prediction. Non-English studies, purely qualitative studies and studies without a clear depression outcome were excluded. The literature informed the predictor selection, analytical framework and hypotheses used in this study.

2.2 EPIDEMIOLOGY OF DEPRESSION IN OLDER ADULTS

Depression is a common mental health condition among older adults. Estimates of prevalence vary according to differences in populations, measurement tools and diagnostic thresholds. Nevertheless, systematic reviews estimate a global prevalence of approximately 28.4% among

adults aged 60 years and above (Hu et al., 2022). Later-life depression is associated with frailty, multimorbidity, functional decline, poorer cognitive function and greater reliance on health and social care services, contributing to increased healthcare costs (Zivin et al., 2013).

Although depression has been widely studied in older adults, much of the existing literature focuses on prevalent rather than incident depression. Prevalence studies are valuable for estimating disease burden and identifying associated factors at a particular point in time but provide limited insight into factors that precede its development. Longitudinal studies of incident depression are therefore important because predictors are measured before the outcome, providing a stronger basis for identifying factors associated with future depression risk (National Institute of Mental Health, 2020).

Identifying depression in older adults presents particular challenges. Depressive symptoms often occur alongside chronic disease, disability, cognitive decline and social vulnerability, which can complicate their recognition (World Health Organization, 2023). Symptoms such as fatigue, poor sleep, reduced appetite and diminished functioning can also overlap with physical illness (Holahan et al., 2010), contributing to underdiagnosis and undertreatment in older populations (Handy et al., 2022). These challenges reinforce the value of identifying individuals at elevated risk before clinical presentation.

Overall, the epidemiological evidence supports investigating incident depression longitudinally and examining predictors across multiple risk domains rather than relying on any single factor.

2.3 Theoretical Models of Depression Risk

Several theoretical frameworks explain the development of depression in later life through interacting biological, psychological and social influences.

The biopsychosocial model remains one of the most influential frameworks. Originally proposed by Engel (1977), it suggests that mental health outcomes arise from interactions between biological processes, psychological characteristics and social circumstances. In older adults, biological factors include chronic disease, pain, disability and neurological changes. Psychological factors include coping ability and resilience, while social factors include loneliness, social participation, financial security and relationships (Tripathi, Das and Kar, 2019). This framework supports the inclusion of multiple predictor domains in depression modelling.

The social determinants of health framework provides a complementary perspective by emphasising socioeconomic conditions and access to resources. Financial hardship, limited social participation and restricted access to support may increase psychological stress and reduce opportunities to maintain mental wellbeing. This is particularly relevant in older adulthood, where retirement, bereavement and declining mobility can reduce income, social contact and independence (Hinnant et al., 2022).

The stress-vulnerability model explains how underlying vulnerability interacts with environmental stressors to influence whether depression develops. Older adults may experience multiple stressors, including chronic illness, functional decline, loss of independence and caregiving responsibilities. These stressors may have greater effects among individuals with higher vulnerability or fewer protective resources (Zubin and Spring, 1977).

Functional decline provides a further pathway linking ageing with depression. Reduced mobility may limit independence and social participation, increasing vulnerability to depression. Functional limitation may therefore act both as a marker of physical deterioration and as a mechanism through which health and social factors influence depression (Yoon & Jee, 2026).

Together, these frameworks support a multidomain approach to depression prediction incorporating sociodemographic, health, behavioural and social factors.

2.4 Multidomain Predictors of Depression in Older Adults

2.4.1 Sociodemographic Predictors

Sociodemographic factors, including age, sex, relationship status and financial situation, are routinely examined as predictors of depression in older adults. Although generally non-modifiable, these characteristics influence exposure to health, behavioural and social risk factors across the life course (Geurgas et al., 2025).

Age has frequently been associated with depression, although findings are mixed. Some studies report higher depression risk among the oldest adults because of increased frailty, chronic disease and loss of independence (Maier et al., 2021). Others suggest that adaptation, resilience or selective survival may weaken the relationship between age and depression (Karel, 1997). Sex differences are more consistent, with women generally experiencing higher rates of depression than men across the lifespan. Proposed explanations include biological factors, social roles and differences in exposure to stress, although these mechanisms remain difficult to separate (World Health Organization, 2025).

Socioeconomic circumstances are also important. Financial hardship has been consistently linked with depression because economic strain can generate chronic stress, reduce access to healthcare and limit opportunities for social participation. In older adults, financial insecurity may be particularly harmful because income may be fixed and opportunities to recover financial stability may be reduced (Hinnant et al., 2022).

Relationship status is another relevant factor. Married or partnered older adults often report lower depression levels than those who are widowed, divorced or living alone, partly because close relationships provide emotional support, companionship and practical assistance. However, relationship quality may be more important than relationship status alone, meaning that simple categorical measures may not fully capture the protective effect of social support (Grundström et al., 2021).

Together, these factors provide important context for understanding differences in depression risk and allow their predictive contribution to be compared with health, behavioural and social factors.

2.4.2 Health Predictors

Physical health is one of the strongest and most consistent determinants of depression in later life. Chronic disease, functional limitation and sleep disturbance can reduce quality of life, restrict independence and increase vulnerability to poor mental health.

Functional limitation has emerged as a particularly important predictor. Chan et al. (2020) reported that mobility impairment was associated with higher depressive symptoms even after adjustment for demographic and clinical factors. Yoon and Jee (2026) similarly found that declining physical function increased depression risk through reduced independence and opportunities for meaningful activity.

Chronic disease burden also contributes to depression risk. Older adults commonly experience multiple long-term conditions, including cardiovascular disease, diabetes, arthritis and respiratory disease. These conditions may increase depression risk through pain, fatigue, reduced activity and uncertainty about future health (Zis et al., 2017). Beurel et al. (2020) also emphasise the bidirectional relationship between physical illness and depression, whereby chronic disease increases depression risk and depression can worsen health through behavioural and physiological mechanisms.

Sleep disturbance is another important health predictor. Poor sleep is common among older adults and has been associated with subsequent depression, although the direction of the

relationship is complex. Sleep disruption may increase emotional vulnerability and impair cognitive function, while depression can also worsen sleep quality. Nevertheless, evidence indicates that sleep disturbance remains associated with depression after adjustment for other health factors (Franzen & Buysse, 2008).

These health factors often coexist with social isolation and financial hardship, meaning their predictive value may reflect both individual associations and relationships across domains.

2.4.3 Behavioural Predictors

Behavioural predictors are particularly relevant because they include potentially modifiable factors such as physical activity, smoking, alcohol consumption and diet.

Physical activity has received the greatest attention. Regular activity is associated with better physical function, reduced inflammation, improved cognitive functioning and greater opportunities for social interaction. Systematic reviews suggest that physically active older adults report fewer depressive symptoms than inactive adults, although effect sizes vary across studies (Mahindru et al., 2023). Variation in measurement, exercise intensity and population characteristics likely contributes to inconsistent findings.

Diet has also been linked with mental health. Higher intake of fruits, vegetables and nutrient-rich foods has been associated with better psychological wellbeing, while diets high in processed foods have been linked with poorer mental health. Molero et al. (2025) reported that healthier dietary patterns were associated with lower depression prevalence across ageing cohorts. However, differences in dietary assessment methods limit direct comparison between studies.

Evidence for smoking and alcohol is more mixed. Smoking is associated with depression in observational research, but confounding by socioeconomic circumstances and physical health is difficult to exclude (Galan et al., 2022). The relationship between alcohol and depression is particularly complex. Heavy use is associated with poorer mental health, while evidence for light or moderate alcohol consumption remains inconsistent (Peter et al., 2022).

Overall, behavioural factors appear to be weaker and less consistent predictors of depression than health or social factors. Previous studies have reported stronger associations for loneliness (adjusted odds ratio [aOR] = 2.33, 95% CI [1.62, 3.34]) and functional limitations (aOR = 1.68 to >8.0), while associations with behavioural factors have been more variable (Abdalla et al., 2026; Meshkat et al., 2025). Nevertheless, behavioural predictors remain

relevant because of their modifiable nature and potential contribution alongside other domains.

2.4.4 Social Predictors

Social connectedness is consistently associated with depression among older adults, with loneliness, social contact and social participation receiving particular attention.

Previous research has consistently identified loneliness as an important predictor of later-life depression. Although loneliness and depression are related, they are conceptually distinct. Loneliness reflects a subjective perception of inadequate social relationships, whereas depression is a broader syndrome involving mood, cognition and behaviour. Longitudinal studies show that loneliness predicts later depression even after adjustment for sociodemographic and health-related factors. However, some overlap in their measurement may contribute to observed associations (Bandyopadhyay, 2024; see Section 5.9).

Social participation may also reduce depression risk. Older adults who maintain regular contact with friends, family and community organisations tend to report better psychological wellbeing (Fernandez-Portero et al., 2023). Participation in organised activities may provide emotional support, identity and purpose. Lorenti et al. (2025) found that volunteering and community engagement were associated with lower depression risk, although the strength of these associations varied by frequency and quality of participation.

Overall, the literature indicates that depression risk in later life spans multiple domains. Health and social predictors generally show the strongest and most consistent associations, while behavioural and sociodemographic factors provide additional information. This evidence supports the multidomain modelling strategy adopted in this study.

2.5 Machine Learning in Mental Health Prediction

The increasing availability of longitudinal health datasets has encouraged the use of machine learning in mental health research. Unlike traditional statistical models, which often focus on estimating associations, machine learning prioritises prediction by identifying patterns across multiple predictors (Nickson et al., 2023).

Machine learning may be particularly useful where relationships between predictors and depression risk are complex. Algorithms such as Random Forest and XGBoost can model interactions and non-linear relationships without requiring them to be specified in advance (Cho et al., 2019). Logistic Regression remains widely used as a benchmark because it is

transparent, interpretable and often performs well when relationships are relatively simple (Shipe et al., 2019).

However, evidence that complex machine learning models outperform simpler methods in mental health prediction is mixed. Tree-based methods can perform well in complex settings, but studies using structured survey data often show only modest improvements over Logistic Regression (Cho et al., 2019). This is particularly relevant for ageing cohorts, where predictors often consist of questionnaire responses and composite scores.

Machine learning also presents methodological challenges. Healthcare datasets often contain missing values, correlated predictors and imbalanced outcomes. Incident depression is particularly difficult to predict because relatively few individuals develop depression during follow-up. Under these conditions, models may favour the majority class and achieve high specificity but poor sensitivity (Dablain et al., 2024). Model evaluation should therefore consider discrimination, sensitivity, specificity and calibration rather than accuracy alone.

2.6 Interpretability in Machine Learning

Interpretability is important when machine learning is applied to healthcare. Clinicians and researchers need to understand how predictions are generated if models are to inform decision-making. Complex models may offer predictive gains but can reduce transparency (Lamba et al., 2026).

Logistic Regression remains valuable because coefficients can be converted into odds ratios (ORs), allowing the direction and magnitude of associations to be interpreted. However, this transparency depends on assumptions about linearity and additive effects (Shipe et al., 2019).

For more complex models, explanation methods are needed. SHapley Additive exPlanations (SHAP) estimate the contribution of each predictor to model predictions, providing both global feature importance and individual-level explanations (Lundberg and Lee, 2017). Permutation importance provides a complementary approach by randomly permuting each predictor and measuring the resulting reduction in model performance. Predictors associated with greater reductions are considered more influential because the model relies more heavily on them for prediction (Chen, 2023).

Together, these approaches provide complementary forms of model interpretation. Accordingly, this study uses Logistic Regression aORs, SHAP and permutation importance.

2.7 Longitudinal Prediction and Temporal Change

Longitudinal designs are valuable because predictors are measured before outcomes occur, reducing concerns about reverse causation. This is particularly important for depression prediction because cross-sectional studies cannot determine whether risk factors preceded depression or arose as a consequence of it (Toh and Hernán, 2008).

Most depression prediction studies use baseline characteristics only. Repeated measurements also allow changes in predictors to be examined. Changes in physical activity, loneliness or alcohol consumption may capture dynamic risk processes that baseline measurements miss.

Evidence for the predictive value of temporal change remains mixed. Some studies show modest benefits from temporal information, while others find little improvement beyond baseline predictors. Gell et al. (2024) suggest that simple change scores may introduce measurement error because they combine uncertainty from repeated measurements. They may also fail to capture longer-term trajectories, cumulative exposure or periods of rapid deterioration. This creates a need to compare baseline and temporal prediction models directly in older cohorts.

2.8 Depression Research Using ELSA

ELSA is one of the most comprehensive longitudinal datasets for studying ageing in England. Since 2002, it has followed adults aged 50 years and older through repeated assessments of physical health, mental health, socioeconomic circumstances and social participation (Steptoe et al., 2012).

Previous ELSA studies have demonstrated associations between depressive symptoms and factors including loneliness, disability and socioeconomic circumstances (Lee et al., 2020; Torres et al., 2016). However, many studies have focused on prevalent symptoms or individual associations rather than predicting incident depression. Fewer studies have applied machine learning while integrating sociodemographic, health, behavioural and social factors within a single predictive framework.

These characteristics make ELSA well suited to investigating incident depression using interpretable machine learning and comparing the predictive contribution of multiple domains.

2.9 Literature Review Summary and Research Gaps

This chapter critically reviewed the literature on depression in older adults, highlighting the contribution of sociodemographic, health, behavioural and social factors. Loneliness, functional limitation, chronic disease burden and poor sleep emerged as consistent predictors, while behavioural factors such as physical activity, diet, smoking and alcohol showed more variable associations. The review also showed that complex machine learning algorithms do not consistently outperform simpler statistical models when applied to structured survey data, while interpretability remains an important consideration in healthcare applications.

Several research gaps were identified. First, much of the existing literature focuses on prevalent rather than incident depression. Second, relatively few studies compare multiple predictor domains within a single predictive framework. Third, evidence remains limited regarding whether temporal changes improve prediction beyond baseline measurements. Finally, few studies combine predictive modelling with complementary interpretability methods.

This study addresses these gaps using longitudinal ELSA data to predict incident depression among adults aged 50 years and older. It compares predictive models and domains, examines temporal changes and subgroup performance, and applies complementary interpretability methods. The following chapter describes the methodology used to conduct these analyses.

3 METHODOLOGY

The previous chapter established the evidence base and research gaps informing this study. This chapter describes the methodology used to address the study aim, research questions and hypotheses. It outlines the study design, data source, sample selection, outcome and predictor variables before describing data preprocessing, model development and evaluation, interpretability methods, additional analyses, ethical considerations and the software and implementation used in the study.

3.1 STUDY DESIGN

This study adopted a quantitative observational research design using secondary data from ELSA. A longitudinal cohort design was used, with predictor variables measured at Wave 7 and depression status assessed at Wave 8. Depression prediction was framed as a supervised binary classification task.

As outlined in Section 1.3, the primary aim was to predict incident depression among older adults using sociodemographic, health, behavioural and social variables. The longitudinal design established temporal ordering between predictors and the outcome, allowing factors associated with subsequent depression onset to be examined.

The study combined machine learning with statistical modelling. Predictive performance and model interpretability were assessed to determine how well incident depression could be predicted and which variables contributed most strongly to prediction.

3.2 DATA SOURCE

As previously described, data were obtained from ELSA, a nationally representative longitudinal cohort of adults aged 50 years and older living in England (Steptoe et al., 2012). Waves 7 and 8 were used. Wave 7 data were collected between June 2014 and May 2015, while Wave 8 data were collected between May 2016 and June 2017 (ELSA, 2018). These waves were selected because they preceded the COVID-19 pandemic, reducing the likelihood that pandemic-related disruption influenced depression outcomes or social behaviours.

Both core interview datasets and derived variable datasets were used. The latter included standardised mental health measures, including the Centre for Epidemiologic Studies Depression Scale (CES-D), which was used to define the outcome variable (American Psychological Association, 2011; see Section 3.4).

3.3 Sample Selection

Participants were eligible if they participated in both Waves 7 and 8 and had valid CES-D scores at both waves. Those meeting the depression threshold at Wave 7 were excluded so that the outcome represented incident depression between Waves 7 and 8 rather than depression already present at baseline. After applying these criteria, the analytical cohort comprised 6,702 participants, all of whom were included in model development.

Incident depression was relatively uncommon within the cohort (see Section 4.1). Class imbalance was therefore considered during model training, validation and evaluation because models trained on imbalanced datasets can favour the majority class (Dablain et al., 2024).

3.4 Outcome Definition

Depression status was measured using the CES-D, a widely used self-report screening instrument for depressive symptoms in population-based studies (American Psychological Association, 2011). The ELSA version consists of eight items assessing symptoms experienced during the previous week, including sadness, loneliness and reduced enjoyment of life. Responses are recorded as binary indicators, with one point awarded for each endorsed symptom. Scores range from 0 to 8, with higher scores indicating greater depressive symptom burden (ELSA, 2018).

A binary depression outcome was created using the established CES-D threshold. Participants scoring four or more were classified as having depression, while those scoring below four were classified as not having depression. Wave 8 depression status served as the prediction target. Because participants with depression at Wave 7 were excluded, the outcome represented transition from non-depressed status at baseline to depression approximately two years later.

However, the outcome did not distinguish between first-onset and recurrent depression. Participants classified as non-depressed at Wave 7 may have experienced depression before baseline (see Section 5.9).

3.5 Predictor Variables

Predictor variables were selected based on evidence reviewed in Section 2.4 and grouped into four domains: sociodemographic, health, behavioural and social. This structure supported interpretability and domain-specific analyses. Table 3.1 summarises the final predictor set.

3.5.1 Sociodemographic Variables

Sociodemographic variables included age, sex, relationship status and financial situation. Age was treated as continuous and sex as binary. Relationship status was treated as a nominal variable with three categories, while financial situation was treated as a six-level ordinal variable.

3.5.2 Health Variables

Health variables included functional limitation, chronic disease burden and poor sleep. Functional limitation was represented by a composite score derived from ten mobility and physical functioning indicators, with higher scores indicating greater limitation. Chronic disease burden was represented by a count of diagnosed long-term conditions. Poor sleep was represented by a binary indicator of restless sleep during the previous week.

3.5.3 Behavioural Variables

Behavioural variables included physical activity, smoking status, alcohol consumption frequency and diet. Physical activity was represented by a composite score derived from vigorous, moderate and mild activity frequencies, with higher values reflecting greater physical activity. Smoking status was represented by a binary indicator of current smoking, while alcohol consumption frequency was retained as an ordinal variable. Diet was represented by a composite score derived from daily fruit and vegetable consumption, with higher scores indicating greater combined intake.

3.5.4 Social Variables

Social variables included loneliness frequency, frequency of meeting friends and social club membership. Loneliness and frequency of meeting friends were treated as ordinal variables, while social club membership was binary. Loneliness conceptually overlaps with the CES-D outcome because the scale includes the item "I felt lonely". Baseline loneliness was retained as a predictor of subsequent depression because the measures were collected at different time points (see Section 5.9). Volunteering frequency was initially considered as a social predictor but was excluded because of substantial missing data (see Section 3.6.2).

Table 3.1

Predictor Variables Included in the Modelling Framework

Domain	Variable	Measurement	Type
Sociodemographic	Age	Age in years (50–90+, with ages ≥90 collapsed in ELSA for disclosure control)	Continuous
	Sex	Male or female	Binary
	Relationship status	Three-category relationship status (married, cohabiting or not partnered)	Nominal
	Financial situation	Six-level self-reported financial management scale (1 = Manage very well, 6 = Severe financial difficulties)	Ordinal
Health	Functional limitation score	Sum of 10 mobility limitation indicators (range: 0–10); higher scores indicate greater functional limitation	Discrete
	Chronic disease count	Count of diagnosed chronic conditions (range: 0–11); higher scores indicate greater disease burden	Discrete
	Poor sleep	Reported restless sleep during the previous week (Yes/No)	Binary
Behavioural	Physical activity score	Sum of reverse-coded vigorous, moderate and mild physical activity frequencies (range: 3–12); higher scores indicate greater overall physical activity	Discrete
	Smoking status	Current smoker (Yes/No)	Binary
	Alcohol consumption frequency	Eight-level drinking frequency scale (1 = Almost every day, 8 = Not at all during the previous 12 months)	Ordinal
	Diet score	Sum of self-reported daily fruit and vegetable portions (range: ≥0); higher scores indicate greater combined intake	Discrete

Domain	Variable	Measurement	Type
Social	Loneliness frequency	Three-level scale (1 = Hardly ever or never, 2 = Some of the time, 3 = Often)	Ordinal
	Frequency of meeting friends	Six-level scale (1 = Three or more times per week, 6 = Less than once per year or never)	Ordinal
	Social club membership	Member of at least one social club (Yes/No)	Binary

Note. Predictor variables were selected based on the literature reviewed in Section 2.4 and grouped into four conceptual domains to support model development, interpretability and domain-specific analyses.

3.6 Data Preprocessing

3.6.1 Data Cleaning

ELSA variable names were converted to lowercase for consistency across datasets. ELSA-specific missing value codes (-9, -8, -2 and -1) were replaced with null values. Wave 7, Wave 8 and derived variable files were merged using the unique participant identifier. Composite variables were created for functional limitation, chronic disease count, physical activity and diet. Physical activity variables were reverse coded before summation so that higher scores reflected greater activity. Categorical and numerical predictors were checked for consistent coding before model development.

3.6.2 Missing Data Handling

Missing data were assessed for all candidate predictors by calculating and visualising the proportion of missing observations for each variable. Missingness was assumed to be missing at random (MAR), whereby missingness depended on observed participant characteristics rather than unobserved values.

Volunteering frequency exhibited a high level of missingness. Including this variable reduced the number of participants with complete predictor data from 5,065 to 2,212 (see Section 4.2). It was therefore excluded from the final feature set, and the robustness of this decision was examined through sensitivity analysis (see Section 4.11). Missing values in the remaining

predictors were handled using Multiple Imputation by Chained Equations, implemented through IterativeImputer in scikit-learn (see Section 3.14).

3.6.3 Feature Encoding and Scaling

Relationship status was one-hot encoded because its categories were nominal. Financial situation was also one-hot encoded to allow category-specific effects without assuming equal differences between adjacent categories. Other ordinal variables were retained as single predictors to preserve their ordering, while binary variables retained binary coding. Numerical predictors were standardised using z-score normalisation to place variables on comparable scales and improve optimisation for Logistic Regression.

Preprocessing was embedded within the modelling pipelines. Imputation and scaling parameters were estimated using training data within each cross-validation fold and then applied to the corresponding validation data (see Section 3.9). This reduced information leakage and supported unbiased performance estimation.

3.6.4 Multicollinearity Assessment

Multicollinearity was assessed using variance inflation factors (VIFs) calculated from the imputed predictor dataset. Initial assessment showed strong collinearity among vigorous, moderate and mild physical activity variables. These were therefore combined into a single physical activity score before modelling.

Some remaining predictors had elevated VIF values (see Section 4.3). However, they represented conceptually distinct constructs identified in the literature and were retained to preserve theoretical coverage across domains. Logistic Regression incorporated regularisation, while Random Forest and XGBoost are generally less sensitive to multicollinearity than traditional regression models (Sztepanacz & Houle, 2024).

3.7 Modelling Approach

Three supervised learning algorithms were trained and compared: Logistic Regression, Random Forest and XGBoost (H1).

3.7.1 Logistic Regression

Logistic Regression served as the baseline interpretable model. It is widely used in clinical prediction research and allows coefficients to be interpreted through ORs and 95% confidence intervals (CIs) (Shipe et al., 2019; see Section 3.11.1).

3.7.2 Random Forest

Random Forest is an ensemble tree-based model that constructs multiple decision trees using bootstrap samples and combines their predictions to reduce variance and improve stability. It can model non-linear relationships and interactions without explicit specification. Class weighting was applied to partially address outcome imbalance (Kern et al., 2019).

3.7.3 XGBoost

XGBoost is a gradient boosting algorithm that sequentially builds trees to correct previous prediction errors. It incorporates regularisation to reduce overfitting and is commonly effective with structured healthcare data. Its compatibility with SHAP also supported model interpretation (Hakkal & Lahcen, 2024; see Section 3.11.2).

3.8 Model Evaluation

Model performance was assessed using discrimination, classification and calibration metrics. The area under the receiver operating characteristic curve (AUROC) was the primary metric because it measures discrimination independently of a fixed classification threshold.

Additional classification metrics included sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV) and the F1 score. Sensitivity (recall) measures the proportion of depression cases correctly identified, which is important in health research because missed cases may delay support or treatment. Specificity measures the proportion of non-cases correctly identified, helping to limit unnecessary follow-up assessments. PPV (precision) indicates the proportion of predicted depression cases that were correctly classified, while NPV indicates the proportion of predicted non-cases that were correctly classified. These metrics are clinically relevant because they reflect the consequences of false-negative and false-positive predictions (Ben-Haim & Dacso, 2024). The F1 score combines precision and recall and is particularly informative under class imbalance.

Calibration was assessed using calibration curves of the XGBoost model and Brier scores. XGBoost was selected for the calibration curve to assess probability estimates from the more complex boosting model. Calibration evaluates whether predicted probabilities correspond to observed outcome frequencies. The Brier score measures the mean squared difference between predicted probabilities and actual outcomes, with lower values indicating better probabilistic predictions. This complemented AUROC by assessing the accuracy of predicted risks rather than only the model's ability to discriminate between individuals with and without

depression. Accurate risk estimates are important in clinical research because predicted probabilities may inform risk stratification and subsequent decision-making (Yang et al., 2022).

3.9 Validation Strategy

Model validation used stratified five-fold cross-validation. Stratification preserved the proportion of depression cases within each fold, accounting for the imbalanced outcome distribution. In each iteration, four folds were used for training and one for validation, with performance metrics averaged across all five folds to provide more robust estimates of model generalisability than a single train-test split (Widodo et al., 2022).

3.10 Hyperparameter Tuning

Hyperparameter optimisation was performed for XGBoost using grid search. Candidate combinations of maximum tree depth and number of estimators were evaluated using AUROC as the optimisation criterion because discrimination was the primary modelling objective. Tuning was performed within the training data to minimise optimistic performance estimates.

3.11 Interpretability Methods

Interpretability analyses were undertaken to identify influential predictors and improve transparency (H5). Three complementary approaches were used: multivariable Logistic Regression, SHAP and permutation importance.

3.11.1 Multivariable Logistic Regression

A multivariable Logistic Regression model was fitted using the same Wave 7 predictors as the machine learning models. aORs and 95% CIs were obtained by exponentiating regression coefficients and Wald confidence intervals. Nagelkerke R^2 was calculated to assess the overall explanatory performance of the model. Statistical significance was defined as $p < .05$. “Male”, “married” and “Manage very well” were the reference categories for sex, relationship status and financial situation, respectively.

3.11.2 SHAP Analysis

SHAP was applied to the XGBoost model to quantify individual predictor contributions (Lundberg & Lee, 2017). Mean absolute SHAP values assessed global feature importance, while summary plots visualised the direction and magnitude of these contributions.

3.11.3 Permutation Importance

Permutation importance was calculated for the Random Forest model by measuring the reduction in model performance after randomly permuting each predictor. Agreement with SHAP rankings was used to assess the robustness of predictor importance (Chen, 2023).

3.12 Additional Analyses

Additional analyses were conducted to address the secondary research questions.

3.12.1 Domain-specific Analysis

Domain-specific XGBoost models were developed using predictors from the four domains: sociodemographic, health, behavioural and social. XGBoost was used to capture potential non-linear relationships. Their predictive performance was compared to assess the contribution of each domain (H2).

3.12.2 Subgroup Analysis

Subgroup analyses examined whether XGBoost performance differed by sex and age group (H4). XGBoost was used consistently to enable comparison across subgroups. Separate models were fitted for males and females and for age groups 50 to 59, 60 to 69, 70 to 79 and 80 years and older.

3.12.3 Temporal Analysis

Temporal analysis examined whether changes in selected predictors improved prediction (H3). Physical activity, loneliness and alcohol consumption were selected as modifiable factors. Change variables were calculated between Waves 7 and 8. Logistic Regression was used for a direct, interpretable comparison of baseline and temporal predictors.

3.12.4 Sensitivity Analysis

Sensitivity analysis assessed the robustness of excluding volunteering frequency because of its substantial missingness. Model performance with and without volunteering frequency was compared to determine whether its exclusion materially affected the results.

3.13 Ethical Considerations

This study was classified as no risk because it involved secondary analysis of robustly anonymised ELSA data with no direct participant contact or new data collection. Ethical approval for original ELSA data collection was obtained by the study investigators (Steptoe et al., 2012). The analysis included sensitive variables, such as depressive symptoms, loneliness

and financial hardship. Findings were therefore interpreted as estimates of depression risk rather than clinical diagnoses and reported only at aggregate level. Institutional ethical approval for this study was obtained from the University of Sheffield before data analysis (Ethics Reference Number: 073342; see Appendix 8.1).

3.14 Software and Implementation

Data preprocessing, statistical analysis and machine learning modelling were conducted in Python 3.13 using Jupyter Notebook within Visual Studio Code. Data manipulation used pandas and NumPy, while visualisations were produced with Matplotlib and Seaborn. Logistic Regression, Random Forest, preprocessing pipelines and cross-validation used scikit-learn, while XGBoost modelling used the XGBoost library. SHAP values were generated with the SHAP library. Statistical analyses, including multivariable Logistic Regression and the calculation of ORs, CIs, *p* values and Nagelkerke R^2, were performed using statsmodels.

3.15 Methodology Summary

This chapter described the methodology used to predict incident depression among older adults using longitudinal ELSA data. Logistic Regression, Random Forest and XGBoost were applied to Wave 7 sociodemographic, health, behavioural and social predictors and evaluated using stratified five-fold cross-validation. Interpretability, domain-specific, temporal, subgroup and sensitivity analyses were also conducted. The following chapter presents the results of these analyses.

4 RESULTS

The previous chapter described the methodology used to develop and evaluate predictive models of incident depression among older adults. This chapter presents the resulting findings. It begins by describing the study sample, missing data and multicollinearity before reporting model performance, predictor importance, domain-specific results, subgroup performance, temporal modelling and sensitivity analysis.

4.1 SAMPLE CHARACTERISTICS

Following dataset merging and eligibility filtering, the analytical cohort comprised 6,702 participants aged 50 years and older who were present in both Waves 7 and 8 and did not meet the depression threshold at baseline. By Wave 8, 483 participants (7.2%) had developed depression, while 6,219 (92.8%) remained free from depression, indicating a marked class imbalance. Tables 4.1 and 4.2 summarise the characteristics of the analytical cohort.

Participants had a mean age of 66.07 years (SD = 11.33), and females represented 54.5% (*n* = 3,652) of the cohort. The mean functional limitation score was 1.46 (SD = 2.20), while the mean chronic disease count was 0.12 (SD = 0.37). Poor sleep was reported by 34.4% (*n* = 2,303) of participants, while 9.5% (*n* = 638) were current smokers.

Most participants reported hardly ever or never feeling lonely (78.0%, *n* = 4,786), while 54.6% (*n* = 3,158) met friends at least once a week. Most participants were married (68.3%, *n* = 4,578), and 78.5% reported managing their finances either very well (47.0%, *n* = 2,968) or quite well (31.5%, *n* = 1,990). Membership of at least one social club was reported by 16.8% (*n* = 1,002).

Table 4.1

Descriptive Characteristics of the Analytical Cohort: Categorical Variables

Variable	**Category**	***n* (%)**
Sex	Male	3,050 (45.5)
	Female	3,652 (54.5)
Relationship status	Married	4,578 (68.3)
	Cohabiting	444 (6.6)
	Not partnered	1,680 (25.1)
Financial situation	Manage very well	2,968 (47.0)
	Manage quite well	1,990 (31.5)
	Get by alright	1,194 (18.9)
	Do not manage very well	58 (0.9)
	Some financial difficulties	81 (1.3)
	Severe financial difficulties	20 (0.3)
Smoking status	Current smoker	638 (9.5)
	Non-smoker	6,064 (90.5)
Poor sleep	Yes	2,303 (34.4)
	No	4,399 (65.6)
Alcohol consumption frequency	Almost every day	842 (13.8)
	Five or six days a week	429 (7.0)
	Three or four days a week	895 (14.6)
	Once or twice a week	1,554 (25.4)
	Once or twice a month	721 (11.8)
	Once every couple of months	461 (7.5)
	Once or twice a year	527 (8.6)
	Not at all in the last 12 months	685 (11.2)
Loneliness frequency	Hardly ever or never	4,786 (78.0)
	Some of the time	1,209 (19.7)
	Often	143 (2.3)

Variable	Category	*n* (%)
Frequency of meeting friends	Three or more times a week	816 (14.1)
	Once or twice a week	2,342 (40.5)
	Once or twice a month	1,643 (28.4)
	Every few months	698 (12.1)
	Once or twice a year	201 (3.5)
	Less than once a year or never	79 (1.4)
Social club membership	Yes	1,002 (16.8)
	No	4,946 (83.2)

Note. Descriptive statistics are presented for the analytical cohort (*N* = 6,702). Categorical variables are reported as frequency, *n* (%). Percentages for categorical variables are based on non-missing observations; therefore, denominators vary across variables.

Table 4.2

Descriptive Characteristics of the Analytical Cohort: Numerical Variables

Variable	Mean (SD)	Measurement Scale
Age (years)	66.07 (11.33)	Years (50–90+, ages ≥90 grouped in ELSA)
Functional limitation score	1.46 (2.20)	0–10; higher scores indicate greater functional limitation
Chronic disease count	0.12 (0.37)	0–11; higher scores indicate greater disease burden
Physical activity score	9.05 (2.30)	3–12; higher scores indicate greater overall physical activity
Diet score	5.28 (2.33)	≥0; Higher scores indicate greater combined fruit and vegetable intake

Note. Descriptive statistics are presented for the analytical cohort (*N* = 6,702). Numerical variables are reported as mean (SD). Measurement scales are provided to aid interpretation. SD = standard deviation.

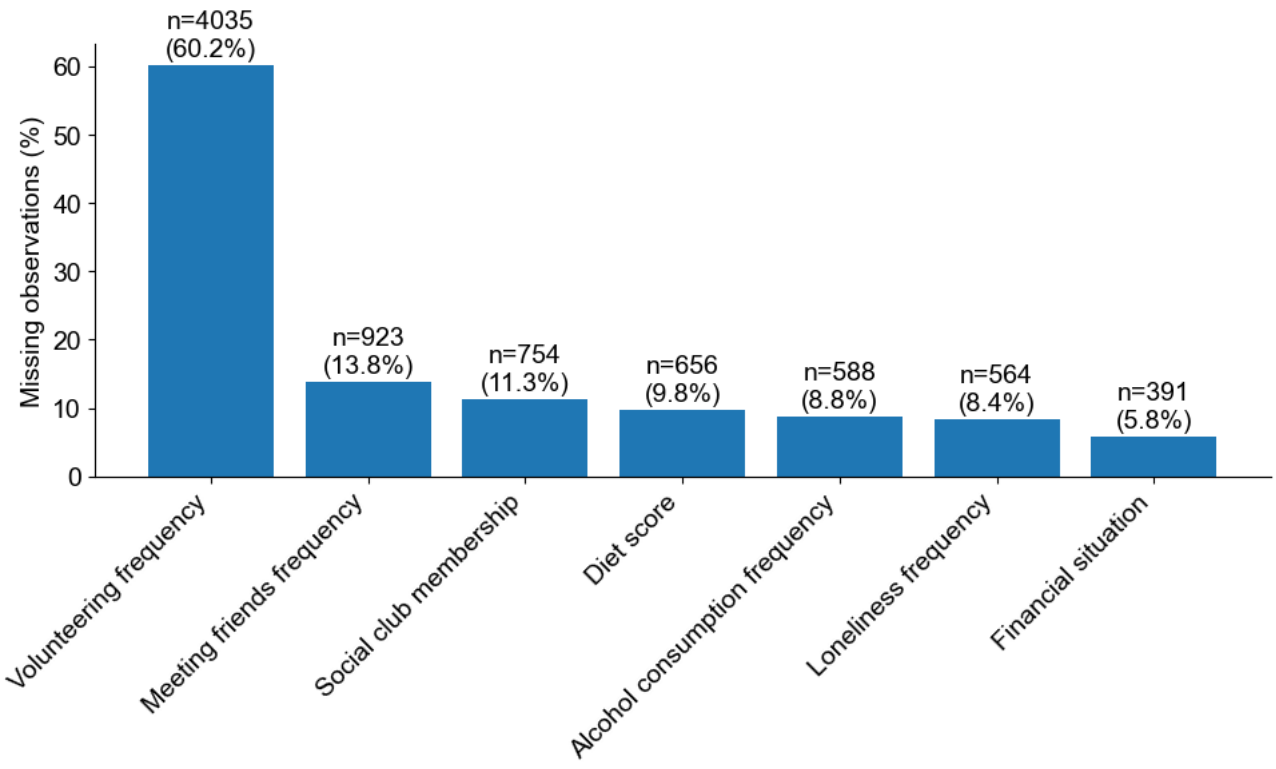
4.2 Missing Data Results

Missingness was generally low, although several social variables showed higher levels. As shown in Figure 4.1 and Table 4.3, volunteering frequency had the highest missingness at 60.2%, substantially exceeding all other variables. Frequency of meeting friends (13.8%) and social club membership (11.3%) also exceeded 10% missingness. Missingness was lower for diet (9.8%), alcohol consumption (8.8%), loneliness (8.4%) and financial situation (5.8%), with little or none among the remaining predictors.

Volunteering frequency was therefore excluded, increasing the number of participants with complete predictor data from 2,212 to 5,065 (75.6%). Sensitivity analysis assessed the effect of this exclusion on model performance (see Section 4.11).

Figure 4.1

Percentage of Missing Data by Predictor Variable



Note. The figure shows the percentage of missing observations before imputation for predictor variables with more than 1% missing data.

Table 4.3

Missing Data by Predictor Variable Before Imputation

Variable	Missing (*n*)	Missing (%)
Volunteering frequency	4,035	60.2
Frequency of meeting friends	923	13.8
Social club membership	754	11.3
Diet score	656	9.8
Alcohol consumption frequency	588	8.8
Loneliness frequency	564	8.4
Financial situation	391	5.8
Physical activity score	3	0.0
Functional limitation score	0	0.0
Chronic disease count	0	0.0
Smoking status	0	0.0
Poor sleep	0	0.0
Relationship status	0	0.0
Sex	0	0.0
Age	0	0.0

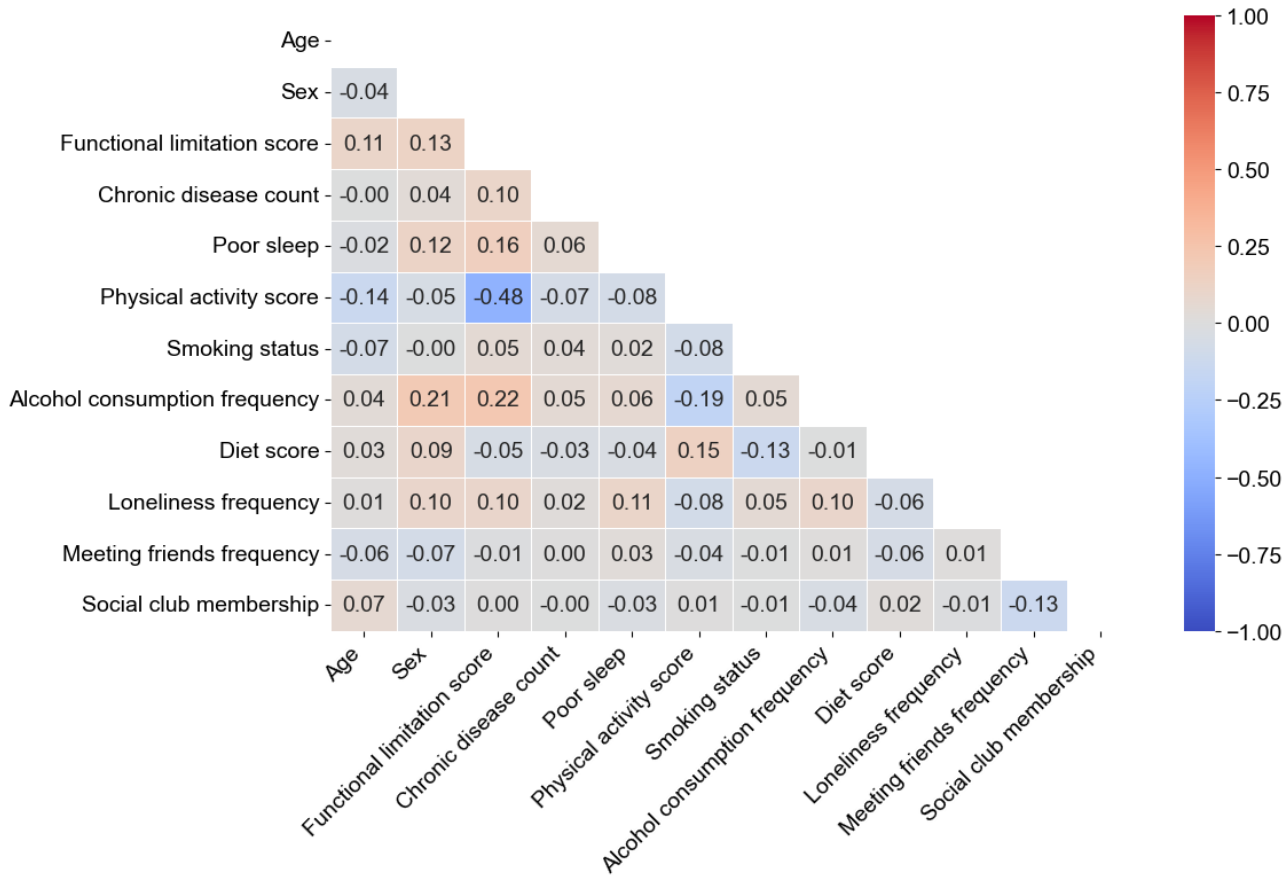
Note. Missing counts (*n*) and percentages are presented before imputation for the analytical cohort (*N* = 6,702). Percentages are rounded to one decimal place.

4.3 Correlation and Multicollinearity

Pairwise correlation analysis was conducted before model development. As shown in Figure 4.2, most predictors were weakly correlated, indicating limited pairwise redundancy. The strongest correlation was between physical activity score and functional limitation score (r = -0.48), with greater physical activity associated with lower functional limitation.

Figure 4.2

Correlation Matrix of Predictor Variables



Note. The figure shows pairwise Pearson correlation coefficients between the predictor variables included in the final modelling framework. Dummy-coded relationship status and financial situation variables were included in the correlation analysis but are omitted from the figure for clarity, as all correlations with the other predictors were below $|r| = 0.30$. Red indicates positive correlations, whereas blue indicates negative correlations. Correlation coefficients closer to 1 or −1 represent stronger linear associations.

VIF analysis provided a further assessment of multicollinearity. As presented in Table 4.4, age had the highest VIF (21.08), followed by physical activity score (15.64) and sex (10.86). Loneliness frequency, frequency of meeting friends, diet score and alcohol consumption frequency also had elevated VIFs, while the remaining predictors had lower values. Despite elevated VIFs for several predictors, pairwise correlations were generally weak. The variables were retained to preserve theoretical coverage of depression risk factors.

Table 4.4

Variance Inflation Factors for Predictor Variables

Variable	VIF
Age	21.08
Physical activity score	15.64
Sex	10.86
Loneliness frequency	8.87
Frequency of meeting friends	7.25
Diet score	6.90
Alcohol consumption frequency	5.98
Functional limitation score	1.93
Poor sleep	1.61
Social club membership	1.26
Smoking status	1.15
Chronic disease count	1.13

Note. VIF values were calculated using the imputed predictor dataset before model development. Relationship status and financial situation dummy variables were included in the analysis but are omitted from the table because all had VIF values below 2, indicating minimal multicollinearity. VIF = variance inflation factor.

4.4 Model Performance

Predictive performance for the three supervised learning models is summarised in Table 4.5. Logistic Regression achieved the strongest discrimination, with an AUROC of 0.759, followed by Random Forest (0.726) and XGBoost (0.668).

Despite moderate discrimination, all models showed poor sensitivity and high specificity at the default classification threshold. Sensitivity ranged from 0.000 for Random Forest to 0.054 for XGBoost, while specificity ranged from 0.985 to 0.999. PPVs and F1 scores were also low, indicating limited identification of participants who subsequently developed depression. Random Forest produced no positive predictions in most cross-validation folds, resulting in mean sensitivity, PPV and F1 scores of zero.

Brier scores were similar across models, with Logistic Regression achieving the lowest value (0.062), compared with Random Forest (0.064) and XGBoost (0.072). Overall, Logistic

Regression demonstrated the strongest predictive performance, combining the highest AUROC with the lowest Brier score.

Table 4.5

Performance of Machine Learning Models

Metric	Logistic Regression	Random Forest	XGBoost
AUROC	0.759	0.726	0.668
Sensitivity	0.027	0.000	0.054
Specificity	0.996	0.999	0.985
PPV	0.310	0.000	0.222
NPV	0.929	0.928	0.931
F1 score	0.049	0.000	0.086
Brier score	0.062	0.064	0.072

Note. Performance metrics represent the mean values obtained across stratified five-fold cross-validation. Lower Brier scores indicate greater accuracy of predicted probabilities. AUROC = area under the receiver operating characteristic curve; PPV = positive predictive value; NPV = negative predictive value.

4.5 Logistic Regression Analysis

Multivariable Logistic Regression was used to estimate aORs and 95% CIs for all predictors. The overall model had a Nagelkerke R^2 of .1571. Table 4.6 presents the adjusted associations, including overall *p* values for multi-category predictors.

Among the social variables, loneliness frequency showed the strongest association with incident depression (aOR = 2.57, 95% CI [2.16, 3.04], $p < .001$). Greater loneliness was associated with substantially higher odds of developing depression during follow-up.

Within the health domain, poor sleep (aOR = 1.69, 95% CI [1.38, 2.06], $p < .001$) and functional limitation score (aOR = 1.18, 95% CI [1.13, 1.23], $p < .001$) were significantly associated with higher odds of incident depression.

Within the behavioural domain, physical activity showed a protective association, with each unit increase in physical activity score associated with lower odds of incident depression (aOR = 0.92, 95% CI [0.88, 0.97], $p < .001$).

Financial situation was the only sociodemographic predictor significantly associated with incident depression overall ($p < .001$). Compared with participants who reported managing financially very well, those who did not manage very well had almost three times greater odds of incident depression (aOR = 2.99, 95% CI [1.49, 6.00], $p = .002$). Participants who reported getting by alright (aOR = 1.62, 95% CI [1.26, 2.08], $p < .001$) or experiencing some financial difficulties (aOR = 1.98, 95% CI [1.04, 3.79], $p = .038$) also had significantly higher odds. Relationship status was not significantly associated with incident depression overall ($p = .997$).

Age, sex, smoking status, alcohol consumption frequency, diet score, chronic disease count, frequency of meeting friends and social club membership were not significantly associated with incident depression after adjustment for the remaining predictors.

Overall, significant associations were identified across the social, health, behavioural and sociodemographic domains, with loneliness showing the largest aOR among the significant predictors.

Table 4.6

Multivariable Logistic Regression Analysis of Incident Depression

Domain	Predictor	aOR	95% CI	*p*
Sociodemographic	Age	1.00	0.99, 1.01	.953
	Female	1.18	0.96, 1.46	.121
	Relationship status			.997
	Cohabiting	1.01	0.66, 1.54	.974
	Not partnered	0.99	0.79, 1.25	.945
	Financial situation			<.001***
	Manage quite well	1.12	0.88, 1.42	.366
	Get by alright	1.62	1.26, 2.08	<.001***
	Do not manage very well	2.99	1.49, 6.00	.002**
	Some financial difficulties	1.98	1.04, 3.79	.038*
	Severe financial difficulties	2.41	0.75, 7.72	.140
Health	Functional limitation score	1.18	1.13, 1.23	<.001***
	Chronic disease count	1.00	0.79, 1.28	.984
	Poor sleep	1.69	1.38, 2.06	<.001***
Behavioural	Physical activity score	0.92	0.88, 0.97	<.001***
	Smoking status	1.19	0.89, 1.60	.235
	Alcohol consumption frequency	1.01	0.96, 1.06	.717
	Diet score	0.97	0.93, 1.01	.159
Social	Loneliness frequency	2.57	2.16, 3.04	<.001***
	Frequency of meeting friends	1.05	0.95, 1.16	.357
	Social club membership	0.86	0.63, 1.16	.311

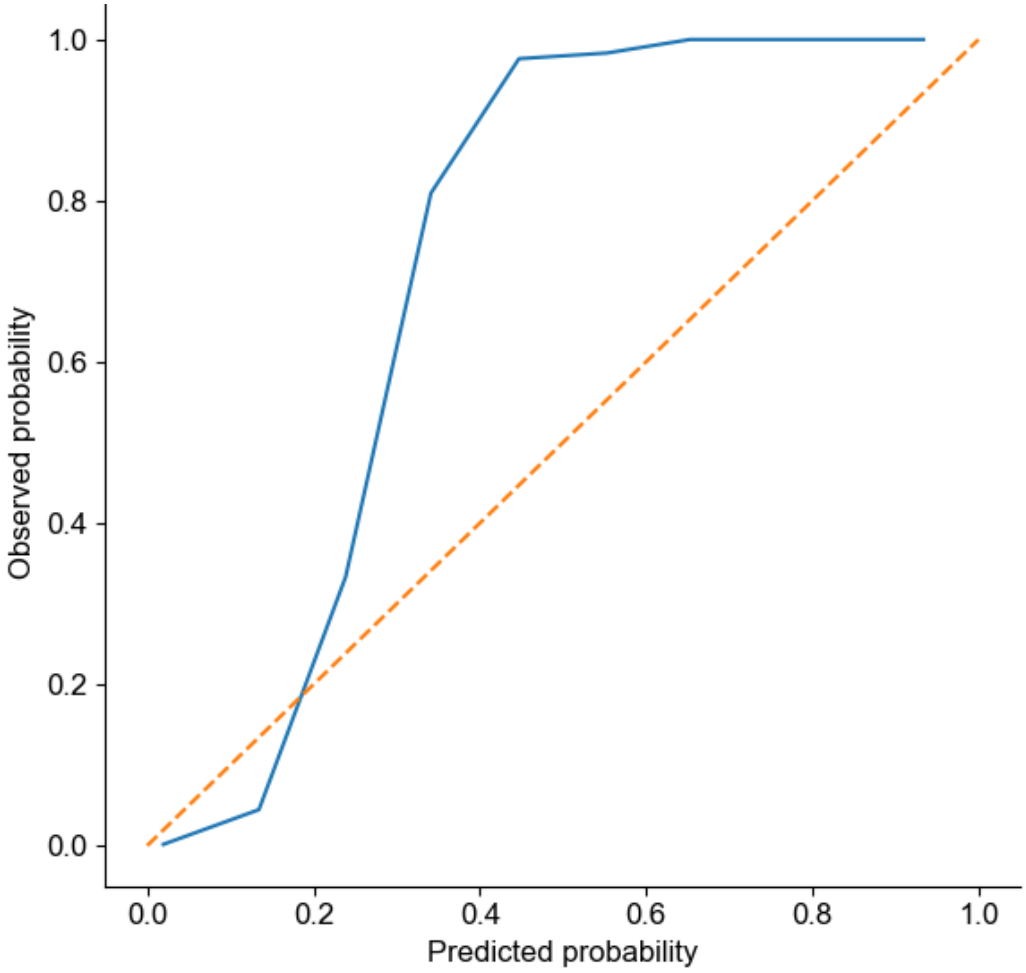
Note. aORs greater than 1 indicate increased odds of incident depression, while values below 1 indicate reduced odds. For relationship status and financial situation, *p* values on the main rows represent likelihood ratio tests, while category-specific values represent comparisons with the reference category. Reference categories were male for sex, married for relationship status and “Manage very well” for financial situation. CI = confidence interval. $p < .05$*, $p < .01$**, $p < .001$***.

4.6 Calibration Results

Calibration of the XGBoost model is shown in Figure 4.3. The calibration curve deviated from the ideal diagonal reference line. At lower predicted probabilities, the model underestimated the observed frequency of depression, while predictions in the mid-probability range tended to overestimate the observed frequency. Greater deviations were also observed at higher predicted probabilities. Overall, the calibration curve indicated variation between predicted probabilities and observed outcome frequencies across the probability range.

Figure 4.3

Calibration Curve for XGBoost



Note. The figure shows predicted probabilities on the x-axis and observed outcome frequencies on the y-axis. The dashed diagonal line represents perfect calibration, where predicted and observed probabilities are equal.

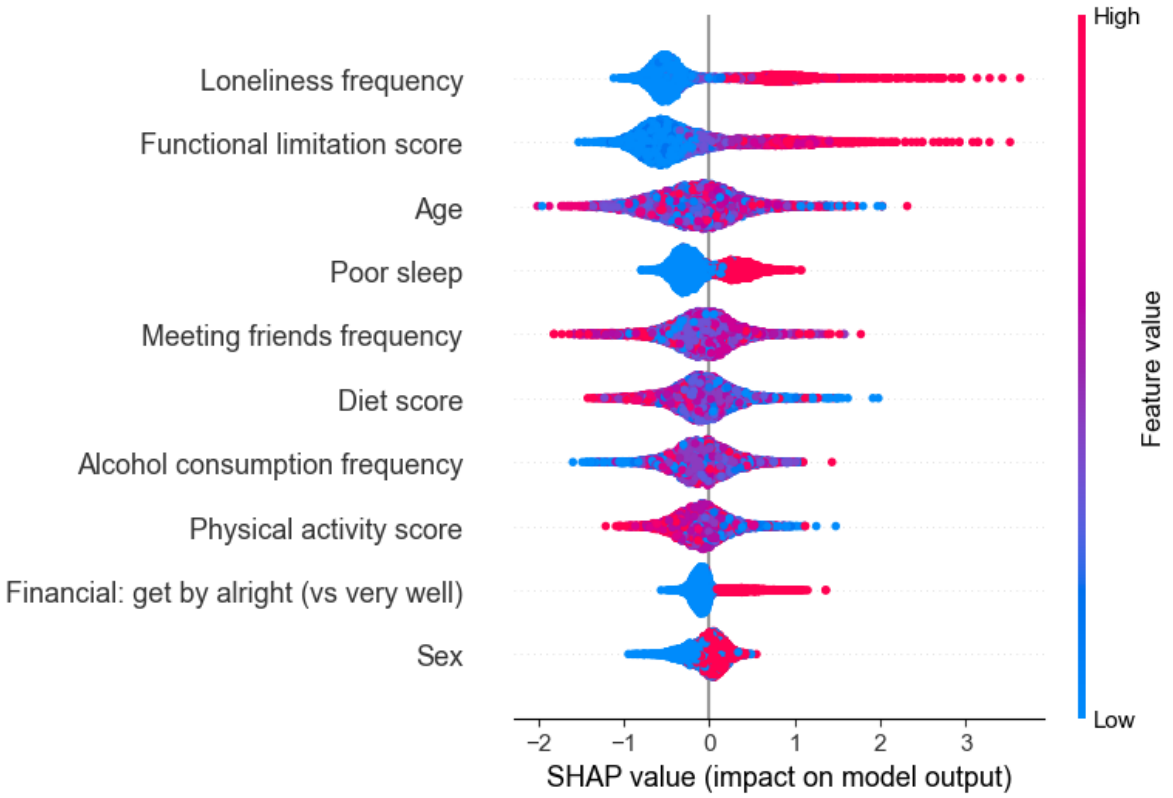
4.7 Feature Importance

4.7.1 SHAP Results

SHAP analysis was used to quantify the contribution of individual predictors to the XGBoost model. The results are presented in Figure 4.4 and Table 4.7. Loneliness frequency had the greatest overall influence on model predictions, followed by functional limitation score, age, poor sleep and frequency of meeting friends. Higher loneliness and greater functional limitation were generally associated with higher predicted probabilities of depression. Poor sleep also contributed positively to predicted risk, while higher physical activity generally reduced it. Age, frequency of meeting friends, diet score and alcohol consumption frequency showed more heterogeneous contributions across participants.

Figure 4.4

SHAP Summary Plot of the Top 10 Predictors



Note. The figure shows SHAP values for the 10 most influential predictors in the XGBoost model. Each point represents an individual participant. Red indicates higher predictor values and blue indicates lower values. Greater distance from zero indicates a larger contribution to the model prediction. Predictors are ranked by mean absolute SHAP value.

Table 4.7

Top 10 Predictors by Mean Absolute SHAP Value

Rank	Feature	Mean Absolute SHAP Value
1	Loneliness frequency	0.589
2	Functional limitation score	0.570
3	Age	0.383
4	Poor sleep	0.295
5	Frequency of meeting friends	0.289
6	Diet score	0.276
7	Alcohol consumption frequency	0.260
8	Physical activity score	0.228
9	Financial: Get by alright (vs Manage very well)	0.156
10	Sex	0.155

Note. Mean absolute SHAP values represent the average magnitude of each predictor's contribution across participants, irrespective of direction. Higher values indicate greater overall influence on model predictions.

4.7.2 Permutation Importance Results

Permutation importance was used to assess the contribution of each predictor to Random Forest model performance. The results are summarised in Table 4.8.

Loneliness frequency and functional limitation score were the two most influential predictors, followed by poor sleep, physical activity score, age and alcohol consumption frequency. The remaining predictors had smaller contributions to model performance.

The rankings were broadly consistent with the SHAP results, particularly for the most influential predictors.

Table 4.8

Top 10 Predictors by Permutation Importance

Rank	Feature	Permutation Importance
1	Loneliness frequency	0.042
2	Functional limitation score	0.035
3	Poor sleep	0.013
4	Physical activity score	0.013
5	Age	0.012
6	Alcohol consumption frequency	0.010
7	Diet score	0.006
8	Financial: Get by alright (vs Manage very well)	0.005
9	Not partnered (vs Married)	0.005
10	Sex	0.004

Note. Permutation importance values represent the mean reduction in model performance following random permutation of each predictor in the Random Forest model. Larger values indicate greater importance for predictive performance.

4.8 Domain-specific Analysis Results

Separate XGBoost models were developed using sociodemographic, health, behavioural and social predictors to compare domain-specific performance. The results are presented in Table 4.9.

Health variables achieved the strongest discrimination (AUROC = 0.686), followed by social variables (AUROC = 0.652). Behavioural (AUROC = 0.572) and sociodemographic variables (AUROC = 0.562) showed weaker discrimination.

Sensitivity was low across all four models, ranging from 0.000 to 0.008, while specificity exceeded 0.99. The health model also achieved the lowest Brier score (0.065).

Overall, health and social variables provided greater predictive performance than behavioural and sociodemographic variables when modelled separately.

Table 4.9

Performance of Domain-Specific Models

Metric	Sociodemographic	Health	Behavioural	Social
AUROC	0.562	0.686	0.572	0.652
Sensitivity	0.008	0.000	0.004	0.002
Specificity	0.996	0.999	0.993	0.998
PPV	0.182	0.000	0.044	0.036
NPV	0.928	0.928	0.928	0.928
F1 score	0.016	0.000	0.007	0.004
Brier score	0.070	0.065	0.070	0.066

Note. Separate XGBoost models were trained using predictors from each conceptual domain. Performance metrics represent mean values across stratified five-fold cross-validation. Lower Brier scores indicate greater accuracy of predicted probabilities. AUROC = area under the receiver operating characteristic curve; PPV = positive predictive value; NPV = negative predictive value.

4.9 Subgroup Analysis Results

XGBoost performance was evaluated across sex and age subgroups to examine variation in prediction. Results are presented in Table 4.10.

Among the sex subgroups, males achieved a higher AUROC (0.675) than females (0.634). Sensitivity was similar for males (0.064) and females (0.065), while specificity exceeded 0.98 in both groups.

Performance also varied across age groups. Participants aged 50 to 59 years achieved the highest AUROC (0.690), followed by those aged 60 to 69 years (0.653), 80 years and older (0.628) and 70 to 79 years (0.620). Sensitivity remained low across all age groups, ranging from 0.029 to 0.069.

Overall, predictive performance varied across sex and age subgroups, although low sensitivity was consistent throughout.

Table 4.10

Performance Across Age and Sex Subgroups

Subgroup	*n*	AUROC	Sensitivity	Specificity	PPV	NPV	F1	Brier
Male	3,050	0.675	0.064	0.989	0.222	0.946	0.098	0.058
Female	3,652	0.634	0.065	0.985	0.282	0.919	0.103	0.083
50–59	1,375	0.690	0.069	0.985	0.267	0.930	0.108	0.074
60–69	2,740	0.653	0.048	0.985	0.157	0.941	0.073	0.064
70–79	1,804	0.620	0.062	0.985	0.323	0.932	0.102	0.072
≥80	621	0.628	0.029	0.962	0.092	0.890	0.042	0.116

Note. XGBoost models were evaluated separately across sex and age subgroups. Performance metrics represent mean values across stratified five-fold cross-validation. Age subgroup sample sizes do not sum to the full analytical cohort because ELSA includes younger partners who fall outside the specified age categories (ELSA, 2018). Lower Brier scores indicate greater accuracy of predicted probabilities. AUROC = area under the receiver operating characteristic curve; PPV = positive predictive value; NPV = negative predictive value.

4.10 Temporal Analysis Results

A Logistic Regression model incorporating change variables was compared with a baseline Logistic Regression model to determine whether changes in predictors improved prediction. Performance metrics are presented in Table 4.11.

The baseline model achieved an AUROC of 0.719, compared with 0.579 for the temporal model. Sensitivity declined from 0.004 to 0.000, while specificity increased slightly from 0.999 to 1.000. The temporal model produced no positive predictions, resulting in an undefined PPV and an F1 score of 0.000. NPV was identical between the models, while Brier scores were similar.

Overall, incorporating change scores between Waves 7 and 8 did not improve predictive performance compared with baseline measurements alone.

Table 4.11

Comparison of Baseline and Temporal Models

Metric	Baseline Model	Temporal Model
AUROC	0.719	0.579
Sensitivity	0.004	0.000
Specificity	0.999	1.000
PPV	0.117	–
NPV	0.928	0.928
F1 score	0.008	0.000
Brier score	0.064	0.066

Note. Logistic Regression was used for both the baseline and temporal models. The baseline model used Wave 7 values for physical activity, loneliness frequency and alcohol consumption frequency, while the temporal model used change scores between Waves 7 and 8 for the same predictors. Performance metrics represent mean values across stratified five-fold cross-validation. PPV was undefined for the temporal model because it produced no positive predictions. AUROC = area under the receiver operating characteristic curve; PPV = positive predictive value; NPV = negative predictive value.

4.11 Sensitivity Analysis Results

A sensitivity analysis examined whether excluding volunteering frequency affected predictive performance. The results are presented in Table 4.12.

AUROC changed only slightly following exclusion of volunteering frequency. The largest reduction occurred for XGBoost (ΔAUROC = −0.011), followed by Random Forest (ΔAUROC = −0.003), while Logistic Regression showed almost no change (ΔAUROC = −0.001).

Overall, excluding volunteering frequency had little effect on model discrimination, supporting its exclusion from the final modelling framework.

Table 4.12

Sensitivity Analysis With and Without Volunteering Frequency

Model	AUROC (With Volunteering)	AUROC (Without Volunteering)	Difference
Logistic Regression	0.760	0.759	−0.001
Random Forest	0.730	0.726	−0.003
XGBoost	0.680	0.668	−0.011

Note. AUROC values are reported for each machine learning model before and after exclusion of volunteering frequency. The difference represents the change in AUROC following removal of the predictor. Negative values indicate a reduction in predictive performance. AUROC = area under the receiver operating characteristic curve.

4.12 Results Summary

This chapter presented the findings of the predictive modelling, interpretability and additional analyses. Table 4.13 summarises the principal results.

Table 4.13

Summary of Key Results

Analysis	Key Finding
Sample characteristics	The analytical cohort included 6,702 participants, of whom 483 (7.2%) developed depression by Wave 8, resulting in substantial class imbalance.
Missing data	Volunteering frequency had the highest missingness (60.2%) and was excluded from the final modelling framework.
Correlation and multicollinearity	Pairwise correlations were generally weak despite some elevated VIFs.
Model performance	Logistic Regression achieved the highest AUROC (0.759), followed by Random Forest (0.726) and XGBoost (0.668).
Logistic Regression	Loneliness, functional limitation, poor sleep and financial hardship were associated with higher odds of incident depression, while physical activity was associated with lower odds.
Calibration	The XGBoost calibration curve showed deviations between predicted probabilities and observed outcome frequencies.
SHAP analysis	Loneliness frequency and functional limitation score were the most influential predictors.
Permutation importance	Loneliness frequency and functional limitation score were again identified as the most important predictors, broadly consistent with the SHAP results.
Domain-specific models	Health (AUROC = 0.686) and social (0.652) models outperformed behavioural (0.572) and sociodemographic (0.562) models.
Subgroup analysis	Predictive performance varied modestly across sex and age groups, although sensitivity remained consistently low.
Temporal modelling	The baseline model outperformed the temporal model (AUROC = 0.719 vs 0.579).
Sensitivity analysis	Excluding volunteering frequency had little effect on AUROC.

The following chapter discusses these findings in relation to the study hypotheses, previous literature and the wider implications for depression prediction among older adults.

5 DISCUSSION

The previous chapter presented the results of the predictive modelling and interpretability analyses. This chapter interprets these findings in relation to the research questions, hypotheses and existing literature. It considers overall model performance, predictor domains, temporal variables and subgroup performance before discussing the most influential predictors, study strengths and limitations, and wider implications.

5.1 INTERPRETATION OF PREDICTIVE PERFORMANCE (H1)

The first hypothesis proposed that models developed using Wave 7 sociodemographic, health, behavioural and social variables would discriminate between participants who did and did not develop depression at Wave 8. This hypothesis was supported.

All three models achieved AUROC values above 0.50, although discrimination varied across models. This did not translate into effective case identification, as sensitivity remained very low while specificity exceeded 98%. The marked class imbalance likely contributed to this pattern, with only 7.2% of participants developing depression during follow-up (*n* = 483/6,702). Models trained under these conditions tend to favour the majority class, resulting in high specificity but poor sensitivity (Dablain et al., 2024).

The fixed classification threshold of 0.50 may also have contributed to the low sensitivity. Although widely used, this threshold is rarely optimal for uncommon clinical outcomes. Lower thresholds would likely increase sensitivity at the expense of specificity. This trade-off may be appropriate for risk stratification, where identifying individuals for further assessment may be more important than minimising false-positive predictions (Rashidi et al., 2023).

Logistic Regression outperformed both Random Forest and XGBoost, showing that greater model complexity did not improve prediction in this study. The structured questionnaire responses and composite scores may have been well suited to linear modelling, while the low event rate may have limited the ability of tree-based models to learn stable non-linear relationships.

This finding is consistent with previous research showing that simpler statistical models often perform as well as, or better than, more complex machine learning methods when applied to structured epidemiological data (Cho et al., 2019). In this study, Logistic Regression therefore provided the strongest predictive performance while retaining greater interpretability.

5.2 Domain Contributions (H2)

The second hypothesis proposed that predictive contribution would differ across the sociodemographic, health, behavioural and social domains. This hypothesis was supported.

The health model achieved the highest discrimination (AUROC = 0.686), followed by the social model (AUROC = 0.652), while behavioural (AUROC = 0.572) and sociodemographic (AUROC = 0.562) models performed less well. Health and social variables therefore provided the greatest standalone predictive performance.

The strong performance of the health domain is consistent with the literature reviewed in Section 2.4.2. Functional limitation and poor sleep were among the strongest predictors identified by Logistic Regression and the interpretability analyses, supporting evidence that declining physical function and sleep disturbance are associated with increased depression risk in older adults. These relationships may also be bidirectional, as depression can contribute to worsening physical function and sleep disturbance over time (Chan et al., 2020; Franzen & Buysse, 2008).

The social domain also demonstrated strong predictive value. Loneliness emerged as the most influential predictor across the interpretability analyses, supporting previous evidence identifying it as an important determinant of depression in later life (Bandyopadhyay, 2024). The strength of this association should be interpreted cautiously because the CES-D outcome includes a loneliness item, creating some conceptual overlap despite loneliness and depression being measured at Waves 7 and 8, respectively (American Psychological Association, 2011).

The behavioural domain produced weaker discrimination. Although physical activity showed a protective association in the Logistic Regression model and ranked highly in the interpretability analyses, smoking, alcohol consumption and diet showed limited predictive importance. Behavioural factors may therefore provide less predictive information than health and social characteristics in this cohort, consistent with previous evidence (Galan et al., 2022; Molero et al., 2025; Peter et al., 2022).

The sociodemographic domain produced the weakest standalone performance. Nevertheless, financial situation remained significantly associated with incident depression in the full Logistic Regression model, suggesting that sociodemographic characteristics provide useful information when considered alongside other domains.

All four domain-specific models performed less well than the full model. This supports the biopsychosocial perspective discussed in Section 2.3 and reinforces the view that later-life depression reflects influences across multiple domains rather than any single group of predictors (Engel, 1977).

5.3 Temporal Variables (H3)

The third hypothesis proposed that incorporating temporal changes in variables between Waves 7 and 8 would improve predictive performance compared with baseline characteristics alone. This hypothesis was not supported.

The temporal model achieved lower discrimination than the baseline model (AUROC = 0.579 vs 0.719) and failed to identify any positive cases at the default classification threshold. Incorporating simple change scores therefore did not improve predictive performance.

Several factors may explain this finding. The two-year interval between survey waves may not adequately capture changes associated with depression development. Change scores also incorporate measurement error from both observations and may introduce additional noise (Stratford & Riddle, 2025). In addition, baseline characteristics such as loneliness, poor sleep and functional limitation may already capture much of the information associated with subsequent depression.

These findings are consistent with previous research suggesting that simple temporal change measures provide limited benefit over baseline characteristics (Gell et al., 2024). They do not suggest that longitudinal information lacks value. More sophisticated approaches, including repeated-measures models, trajectory analysis and survival modelling, may better capture changes in depression risk over time (Nemesure et al., 2024).

5.4 Subgroup Differences (H4)

The fourth hypothesis proposed that predictive performance would differ across age and sex subgroups. This hypothesis was supported, although the differences were modest.

Performance was slightly better among males than females, with AUROC values of 0.675 and 0.634, respectively. Sensitivity remained similarly low in both groups, while specificity exceeded 98%. One possible explanation is that determinants of later-life depression differ between males and females, with some risk factors exerting stronger or more complex effects in women (Li et al., 2022). The lower discrimination among women may therefore reflect relevant factors not captured by the predictors included in this study.

Performance also varied across age groups. The highest discrimination was observed among participants aged 50 to 59 years (AUROC = 0.690), followed by those aged 60 to 69 years (0.653), 80 years and older (0.628) and 70 to 79 years (0.620). Discrimination was therefore generally lower among the older age groups. Greater multimorbidity, frailty and social complexity in later life may increase heterogeneity in depression risk and make prediction more difficult among older participants (Zhang et al., 2025).

Overall, predictive performance varied modestly across age and sex subgroups, while low sensitivity was consistent throughout. External validation is required to determine whether these patterns persist in other populations.

5.5 Model Calibration

Calibration analysis provided additional insight into the XGBoost model beyond discrimination. The calibration curve (Figure 4.3) deviated from the ideal reference line, indicating imperfect agreement between predicted probabilities and observed outcome frequencies. The model underestimated risk at lower predicted probabilities and overestimated risk across parts of the mid-probability range. At higher probabilities, the curve became increasingly unstable, suggesting that some probability estimates were more extreme than supported by the observed data.

These findings indicate that although XGBoost demonstrated some discriminatory ability, its predicted probabilities should be interpreted cautiously. Accurate calibration is particularly important when predicted probabilities are intended to support risk assessment or clinical decision-making. Previous research suggests that post-hoc methods such as Platt scaling and isotonic regression can improve probability estimates in machine learning models (Odesola et al., 2025). Applying these approaches may improve the calibration and practical utility of future depression prediction models.

5.6 Interpretation of Logistic Regression Findings

The Logistic Regression model complemented the machine learning analyses by providing interpretable estimates of associations between individual predictors and incident depression. The overall model had a Nagelkerke R^2 of .1571, indicating that the predictor set provided a modest degree of explanatory fit.

Loneliness showed a strong independent association with incident depression (aOR = 2.57), with greater loneliness associated with substantially higher odds of developing depression. This

finding is consistent with evidence identifying perceived social isolation as an important determinant of depression among older adults. Unlike objective measures of social contact, loneliness reflects an individual's subjective experience of social connectedness, which may contribute to its strong association with subsequent depression (Bandyopadhyay, 2024).

Financial situation also showed a clear pattern, with poorer financial circumstances generally associated with greater odds of depression compared with managing financially very well. This supports the social determinants of health framework, which proposes that financial hardship contributes to chronic stress, reduced access to resources and poorer mental wellbeing (Hinnant et al., 2022).

Within the health domain, poor sleep and functional limitation were independently associated with higher depression risk. These findings align with previous research showing that declining physical function and sleep disturbance are associated with subsequent depression in later life (Chan et al., 2020; Franzen & Buysse, 2008). These factors may also reflect broader deterioration in physical health.

Physical activity showed a significant protective association, with higher activity levels associated with lower odds of developing depression. This supports evidence linking regular physical activity with psychological wellbeing through physiological, behavioural and social mechanisms (Mahindru et al., 2023).

In contrast, age, sex, smoking status, alcohol consumption, chronic disease count, frequency of meeting friends and social club membership were not significant after adjustment for the remaining predictors. Their associations with depression may therefore overlap with stronger social, functional and socioeconomic factors included in the model.

Overall, the Logistic Regression findings emphasised the importance of loneliness, financial circumstances, functional health, poor sleep and physical activity within the multidomain predictor set.

5.7 Feature Importance and Interpretability (H5)

The fifth hypothesis proposed that loneliness, functional limitation and age would emerge among the most influential predictors of depression. This hypothesis was supported.

Across both SHAP analysis and permutation importance, loneliness and functional limitation ranked as the two most influential predictors. Age also ranked among the leading predictors,

placing third in SHAP and fifth in permutation importance. Poor sleep, physical activity, financial situation and alcohol consumption also contributed to model predictions.

The broad agreement between SHAP and permutation importance increases confidence in the predictor rankings because the methods assess importance differently. SHAP quantifies predictor contributions to model predictions, while permutation importance measures the reduction in model performance after random permutation of each predictor. Several of the leading predictors were also significantly associated with incident depression in the Logistic Regression analysis, providing further consistency across the interpretability methods.

The SHAP analysis also provided insight into the direction of predictor contributions. Higher loneliness and greater functional limitation were generally associated with higher predicted depression risk, while higher physical activity was associated with lower predicted risk. Age and alcohol consumption showed more heterogeneous contributions across participants.

5.8 Strengths of the Study

This study has several strengths.

First, it used ELSA, a large, nationally representative longitudinal cohort containing comprehensive information on the health, social and socioeconomic circumstances of older adults in England (Steptoe et al., 2012). The longitudinal design established temporal ordering between predictors measured at Wave 7 and incident depression assessed at Wave 8.

Second, the study focused on incident rather than prevalent depression. Excluding participants with depression at baseline enabled the analysis to identify factors associated with depression onset, providing greater relevance for prevention and early identification than cross-sectional studies (National Institute of Mental Health, 2020).

A further strength was the multidomain modelling framework. Sociodemographic, health, behavioural and social variables were examined simultaneously, allowing comparison of their predictive contributions within a single analytical framework.

The study also employed a comprehensive evaluation strategy. Logistic Regression, Random Forest and XGBoost were compared using discrimination, classification and calibration metrics. Stratified five-fold cross-validation reduced the risk of optimistic performance estimates and improved the robustness of model evaluation (Widodo et al., 2022).

Interpretability represented another strength. Logistic Regression, SHAP analysis and permutation importance provided complementary perspectives on predictor importance, increasing confidence in the identified predictors and improving transparency.

Finally, sensitivity analysis showed that excluding volunteering frequency substantially increased the number of participants with complete predictor data while having minimal impact on predictive performance. This supported its exclusion from the final modelling framework.

5.9 LIMITATIONS

Despite these strengths, several limitations should be considered when interpreting the findings.

First, depression was identified using CES-D scores rather than clinical diagnosis, creating potential for outcome misclassification. Because depression status was assessed only at Waves 7 and 8, participants classified with incident depression may also have experienced depressive episodes before baseline. The outcome therefore represents new-onset depression during the study period rather than first lifetime onset. In addition, the CES-D includes a loneliness item, which may have contributed to the strong association observed between baseline loneliness and subsequent depression despite the measures being collected at different waves (American Psychological Association, 2011).

Second, substantial class imbalance affected model performance. Only 7.2% of the analytical cohort (*n* = 483/6,702) developed depression during follow-up, contributing to consistently low sensitivity across the models. Although discrimination was moderate, the models therefore had limited ability to identify participants who subsequently developed depression.

Third, the default classification threshold of 0.50 was retained throughout model evaluation. Alternative thresholds may have increased sensitivity and may be more appropriate when identifying individuals for further assessment is prioritised over minimising classification error (Rashidi et al., 2023).

Missing data also presented limitations. Volunteering frequency was excluded because of extensive missingness, meaning that one aspect of social participation was not represented in the final models (Lorenti et al., 2025). Iterative imputation was used for the remaining missing data, but violations of the MAR assumption may have introduced bias, particularly if

missingness was related to unobserved characteristics, such as poorer health, greater functional limitations or socioeconomic disadvantage (Pedersen et al., 2017).

The temporal analysis also requires cautious interpretation because change variables incorporated measurements from Wave 8, when the depression outcome was assessed. Unlike the baseline predictors, these variables therefore did not strictly precede the outcome and should not be interpreted as prospective predictors of Wave 8 depression.

Combining vigorous, moderate and mild physical activity into a single score simplified the modelling framework but removed information about activity intensity. Residual multicollinearity also remained among several predictors, although theoretically distinct variables were retained to preserve conceptual coverage.

Finally, external validation was not performed. Although internal cross-validation provided estimates of model performance within ELSA, evaluation in independent cohorts is required to determine the generalisability of the findings.

5.10 Implications for Practice and Research

The findings have implications for clinical practice and future research.

From a clinical perspective, self-reported measures of loneliness, functional health, sleep and financial wellbeing may help identify older adults at elevated risk of depression. Although the models lacked sufficient sensitivity for standalone screening, they may have value for population-level risk stratification and targeted further assessment.

The findings also highlight the relevance of social and functional factors to depression prevention. Loneliness and functional limitation were particularly influential, while poor sleep and financial hardship were also associated with incident depression. These factors may therefore help inform approaches to identifying individuals who could benefit from preventative support.

Methodologically, greater algorithmic complexity did not improve predictive performance. Logistic Regression outperformed Random Forest and XGBoost while retaining greater interpretability, supporting the use of transparent models in healthcare research where explanation is important.

Future research should investigate threshold optimisation, cost-sensitive learning and approaches to class imbalance, including Synthetic Minority Over-sampling Technique (SMOTE), to improve case identification (Chawla et al., 2002). Studies should also exploit the

longitudinal structure of ELSA through repeated-measures modelling, trajectory analysis or survival analysis. External validation in independent ageing cohorts and inclusion of additional predictors, such as cognitive function, medication use and body mass index, could further assess and improve predictive performance.

5.11 Discussion Summary

This chapter interpreted the findings in relation to the research questions, hypotheses and existing literature. The findings support the importance of considering health, behavioural, social and sociodemographic factors together when predicting incident depression in older adults. Moderate discrimination was accompanied by low sensitivity, while temporal change variables provided no additional predictive benefit and subgroup differences were modest. Interpretability analyses provided further insight into influential predictors, while the study's limitations highlighted areas requiring caution and further research. The following chapter concludes the dissertation by summarising the main contributions and recommendations for future research.

6 CONCLUSION

The previous chapter interpreted the study findings and their implications. This final chapter draws together the main conclusions, summarises the contribution and limitations of the research, and outlines recommendations for future work.

6.1 SUMMARY OF THE STUDY

This study investigated incident depression prediction among adults aged 50 years and older using longitudinal ELSA data. Participants free from depression at Wave 7 were followed to Wave 8, using sociodemographic, health, behavioural and social characteristics as predictors. Logistic Regression, Random Forest and XGBoost were evaluated, while aORs, SHAP and permutation importance were used to examine predictor importance. Additional analyses assessed predictor domains, temporal change variables and performance across age and sex subgroups.

6.2 KEY FINDINGS

The findings addressed each of the five research questions. Health variables provided the strongest standalone predictive performance, followed by social variables, while behavioural and sociodemographic domains performed less well.

Temporal change variables did not improve predictive performance compared with baseline characteristics, indicating limited value from the simple change measures examined.

Predictive performance varied modestly across age and sex subgroups. Although differences in discrimination were observed, low sensitivity remained consistent across subgroups, indicating that difficulties identifying incident depression were not confined to a particular demographic group.

Loneliness and functional limitation were the most influential predictors across SHAP and permutation importance analyses and were also associated with incident depression in the Logistic Regression model. Poor sleep, financial hardship and physical activity were also identified as relevant predictors, providing broad consistency across the analytical approaches.

Logistic Regression achieved the strongest overall discrimination (AUROC = 0.759), outperforming Random Forest and XGBoost. Greater algorithmic complexity therefore offered

no predictive advantage in this dataset. Despite moderate discrimination, all models demonstrated low sensitivity and high specificity, limiting their suitability for identifying individuals who subsequently developed depression.

6.3 Contribution and Implications

This study contributes to the literature by focusing on incident depression and evaluating sociodemographic, health, behavioural and social predictors within a common longitudinal modelling framework. It also assessed the predictive value of temporal changes and combined multiple interpretability approaches to examine predictor importance.

The findings support theoretical frameworks that conceptualise later-life depression as multidimensional, including the biopsychosocial model and social determinants of health framework discussed in Section 2.3.

Methodologically, the findings support the use of interpretable approaches for structured epidemiological data. Logistic Regression outperformed Random Forest and XGBoost while providing directly interpretable estimates of association, showing that greater model complexity does not necessarily improve prediction.

Although the models lacked sufficient sensitivity for clinical screening, the findings identify characteristics associated with incident depression and the predictor domains containing the greatest predictive information. This evidence may inform the development of future risk-stratification approaches.

6.4 Limitations and Future Research

Several limitations affect the interpretation and generalisability of the findings. First, only 483 of 6,702 participants (7.2%) developed incident depression, resulting in substantial class imbalance and contributing to low sensitivity. Future research should investigate threshold optimisation, cost-sensitive learning and resampling approaches such as SMOTE to improve case identification while considering effects on specificity and calibration.

Second, depression was measured using the CES-D rather than clinical diagnosis. The outcome therefore reflects depressive symptoms according to the applied threshold rather than clinically diagnosed depressive disorder. Future research should assess whether the findings generalise to clinical diagnoses or alternative validated measures.

Third, temporal analyses were limited to two waves and a small number of repeatedly measured predictors. Change variables also incorporated Wave 8 measurements, when the depression outcome was assessed, limiting their interpretation as prospective predictors. Future studies using additional ELSA waves could examine longer-term trajectories and ensure temporal separation between predictors and outcomes.

Loneliness was included as a baseline predictor while the CES-D outcome contains an item assessing loneliness. Although measured at different time points, this conceptual overlap may have contributed to its strong association with incident depression. Future analyses could assess the robustness of this finding using alternative outcome definitions.

Finally, external validation was not undertaken. Validation in independent ageing cohorts is required to determine whether predictive performance generalises to populations with different demographic, health and socioeconomic characteristics.

6.5 Final Conclusion

This study achieved moderate discrimination in predicting incident depression among older adults using longitudinal population-based data, although low sensitivity limited practical application. Health and social characteristics provided the greatest predictive information, with loneliness and functional limitation emerging as the most influential predictors. Temporal change variables did not improve prediction, subgroup differences were modest, and Logistic Regression outperformed the more complex models.

Overall, the study supports the use of interpretable, multidomain approaches to incident depression prediction in older adults. Improving sensitivity, incorporating richer longitudinal information and externally validating predictive models are key priorities for developing more clinically useful approaches.

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8 APPENDICES

8.1 ETHICS APPROVAL LETTER



Downloaded: 26/03/2026

Approved: 26/03/2026

Uche Dumzo-Ajufo

Registration number: 250124837

School of Information, Journalism and Communication

Programme: MSc Data Science

Dear Uche

PROJECT TITLE: Interpretable Machine Learning for Longitudinal Prediction of Depression in Older Adults: Evidence from Pre-COVID ELSA Waves 7 and 8

APPLICATION: Reference Number 073342

This letter confirms that you have signed a University Research Ethics Committee-approved self-declaration to confirm that your research will involve only existing research, clinical or other data that has been robustly anonymised. You have judged it to be unlikely that this project would cause offence to those who originally provided the data, should they become aware of it.

As such, on behalf of the University Research Ethics Committee, I can confirm that your project can go ahead on the basis of this self-declaration.

If during the course of the project you need to deviate significantly from the above-approved documentation please inform me since full ethical review may be required.

Yours sincerely

Jared Ahmad

Departmental Ethics Administrator

8.2 AI DECLARATION

Generative artificial intelligence (AI) was used during the preparation of this dissertation. ChatGPT (OpenAI) was used as a writing and programming support tool to assist with the following tasks:

- Proofreading sections and suggesting improvements to grammar, clarity and academic writing.
- Suggesting headings and improvements to dissertation structure.
- Debugging and refining Python code used for data preparation, statistical analysis and machine learning.
- Presenting results, including converting statistical outputs into dissertation-ready tables.
- Providing guidance on dissertation formatting and presentation.

ChatGPT supported sections throughout the dissertation. All AI-generated suggestions, code and outputs were reviewed and verified before use. All analyses, interpretations and conclusions are my own, and I take responsibility for the accuracy, originality and integrity of the final submission.

Examples of prompts used during the preparation of this dissertation include:

- "Proofread this section and improve the grammar and academic writing while preserving the original meaning."
- "Suggest clearer section headings and subheadings for this chapter."
- "Help identify and fix errors in this Python code."
- "Convert these results into a dissertation-ready table and improve its presentation."
- "Review this section for consistency with UK English and APA 7 style."

8.3 RESEARCH DIARY AND REFLECTION

8.3.1 Research Diary

8.3.1.1 Supervision Meeting 1: 25 February 2026

- Work completed:
    - Developed an initial dissertation idea.
    - Explored the feasibility of using the ELSA dataset.
- Meeting contents:
    - Refined the project into a longitudinal study using multiple ELSA waves rather than a cross-sectional design.
    - Recommended including subgroup analyses.
    - Discussed literature searching and the ethics process.
- Planned actions:
    - Redesign the study as a longitudinal analysis.
    - Begin the literature review and ethics application.
    - Incorporate subgroup analyses into the study design.

8.3.1.2 Research Proposal Feedback: 2 April 2026

- Work completed:
    - Submitted the research proposal.
- Feedback received:
    - Report PPV and NPV instead of precision and recall.
    - Define abbreviations on first use and improve academic writing through clearer terminology and sentence structure.
    - Expand the literature review through broader searches and more detailed discussion of previous studies.
    - Clarify that the study predicts incident rather than prevalent depression.
    - Explain the interpretability of Random Forest and XGBoost.
    - Assess missing data before model development.
- Planned actions:
    - Revise the proposal in response to the feedback.

8.3.1.3 Supervision Meeting 2: 19 May 2026

- Work completed:
    - Revised the proposal.

    - Began coding and drafting the dissertation.
- Meeting contents:
    - Reviewed progress on the analysis and dissertation.
    - Recommended reporting aORs with 95% CIs.
    - Discussed limitations relating to participants' potential history of depression and missing data.
- Planned actions:
    - Expand the discussion of study limitations.
    - Improve statistical reporting.

8.3.1.4 Supervision Meeting 3: 23 June 2026

- Work completed:
    - Submitted draft Methodology and Results chapters.
- Meeting contents:
    - Recommended adding chapter introductions and summaries to improve flow.
    - Clarified methodological decisions, including CES-D scoring.
    - Investigated the impact of missing data, particularly volunteering frequency, and discussed combining highly related variables where appropriate.
    - Recommended improvements to descriptive statistics, participant reporting, correlation heatmaps and descriptive tables.
    - Recommended presenting complete Logistic Regression results, including reference categories.
    - Recommended expanding interpretability analyses and standardising formatting.
- Planned actions:
    - Remove volunteering frequency and perform a sensitivity analysis.
    - Engineer composite predictor variables and add a sociodemographic domain.
    - Revise the chapters in response to the remaining feedback.

8.3.1.5 Supervision Meeting 4: 21 July 2026

- Work completed:
    - Submitted draft Introduction, Discussion and Conclusion chapters.
- Meeting contents:
    - Recommended mentioning broader determinants of depression, including environmental and biological factors.

    - Recommended reporting sample sizes and proportions consistently.
    - Recommended interpreting findings more critically rather than simply describing the results.
    - Discussed the bidirectional relationship between sleep disturbance and depression and the interpretation of subgroup differences.
    - Clarified the interpretation of model discrimination and class imbalance.
- Planned actions:
    - Revise the Introduction, Discussion and Conclusion chapters.
    - Improve interpretation of findings.
    - Strengthen consistency of reporting.

8.3.1.6 Supervision Meeting 5: 5 August 2026

- Work completed:
    - Submitted a completed dissertation draft.
- Meeting contents:
    - Discussed the overlap between baseline loneliness and the CES-D loneliness item and its inclusion as a study limitation.
    - Recommended adding Nagelkerke R^2 and overall p values for multi-category predictors to the Logistic Regression results.
    - Reviewed the presentation and formatting of tables and figures.
- Planned actions:
    - Revise the dissertation in response to the feedback before final submission.

8.3.2 Reflection on Research

8.3.2.1 Supervisor Feedback 1: Handling Missing Data

Reflection on feedback: My supervisor questioned whether volunteering frequency should remain in the analysis because it contained a high proportion of missing data and greatly reduced the number of participants with complete predictor data. Rather than removing the variable immediately, I investigated its impact by comparing model performance before and after exclusion using a sensitivity analysis. Removing the variable increased the number of participants with complete predictor data from 2,212 to 5,065 while having only a minimal effect on predictive performance. This provided evidence that excluding the variable improved the robustness of the analysis without compromising model accuracy. The experience strengthened my understanding of missing data and the importance of supporting methodological decisions with empirical evidence rather than assumptions.

Associated skills: Research and critical thinking, applying knowledge

Supervisor Feedback 2: Feature Engineering and Predictor Selection

Reflection on feedback: My supervisor suggested reducing redundancy within the predictor set by combining related variables and improving the organisation of the modelling framework. I reviewed how predictors were represented, combined the physical activity variables into a composite score, retained derived measures such as functional limitation score and chronic disease count, and reorganised predictors from three to four conceptual domains: sociodemographic, health, behavioural and social. These changes reduced redundancy, improved the clarity of the descriptive analyses and strengthened the comparison of predictor domains. This process improved my understanding of feature engineering as part of research design and developed my ability to justify analytical decisions using theoretical and statistical reasoning.

Associated skills: Applying knowledge, digital capability

8.3.2.2 Supervisor Feedback 3: Interpreting Results

Reflection on feedback: My supervisor highlighted that several sections of the discussion summarised the results without explaining why the observed patterns may have occurred. In response, I rewrote large parts of the discussion to provide more critical interpretation. For example, I explored possible explanations for differences in predictive performance across age and sex groups, acknowledged the bidirectional relationship between sleep disturbance and depression, and adopted more cautious language when discussing model performance. These revisions produced a more analytical discussion that engaged more effectively with the literature rather than simply restating the results. Through this process, I developed stronger critical thinking skills and a greater appreciation of the importance of interpreting findings within their wider clinical and methodological context.

Associated skills: Academic skills, research and critical thinking

8.3.2.3 Supervisor Feedback 4: Improving Dissertation Structure

Reflection on feedback: My supervisor suggested strengthening links between chapters and improving the overall structure of the dissertation. I responded by adding introductory and summary sections to each chapter, improving transitions, introducing cross-references where appropriate and standardising terminology. I also revised headings, figure titles and table captions for consistency. These changes created a clearer progression from the literature

review to the conclusions. This experience strengthened my academic writing skills and understanding of how effective structure supports the communication of complex research.

Associated skills: Academic skills, personal development

8.3.2.4 Supervisor Feedback 5: Presenting and Interpreting Statistical Findings

Reflection on feedback: My supervisor recommended presenting statistical findings in greater detail and improving the explanation of their clinical relevance. In response, I expanded the Logistic Regression results to report ORs with CIs, improved the interpretation of SHAP and permutation importance analyses, standardised the presentation of tables and figures, and reported participant numbers alongside percentages where appropriate. I also refined the discussion of discrimination, calibration and class imbalance to provide a more balanced interpretation of model performance. These revisions strengthened the transparency of the dissertation and improved my ability to communicate statistical evidence accurately and responsibly.

Associated skills: Academic skills, applying knowledge

8.3.2.5 Evidence of mySkills Reflections

The screenshot below provides evidence that all five reflections were completed and recorded in mySkills in accordance with the dissertation requirements:

